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Counterfactual learning for higher-order relation prediction in heterogeneous information networks

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ABSTRACT

Heterogeneous Information Networks (HINs) play a crucial role in modeling complex social systems, where predicting missing links/relations is a significant task. Existing methods primarily focus on pairwise relations, but real-world scenarios often involve multi-entity interactions. For example, in academic collaboration networks, an interaction occurs between a paper, a conference, and multiple authors. These higher-order relations are prevalent but have been underexplored. Moreover, existing methods often neglect the causal relationship between the global graph structure and the state of relations, limiting their ability to capture the fundamental factors driving relation prediction. In this paper, we propose HINCHOR, an end-to-end model for higher-order relation prediction in HINs. HINCHOR introduces a higher-order structure encoder to capture multi-entity proximity information. Then, it focuses on a counterfactual question: "If the global graph structure were different, would the higher-order relation change?" By presenting a counterfactual data augmentation module, HINCHOR utilizes global structure information to generate counterfactual relations. Through counterfactual learning, HINCHOR estimates causal effects while predicting higher-order relations. The experimental results on four constructed benchmark datasets show that HINCHOR outperforms existing state-of-the-art methods.

1. Introduction

Many real-world scenarios can be effectively represented using graph-structured data (i.e., networks), where entities and their relations are depicted as nodes and edges, respectively. Homogeneous networks consist solely of nodes and edges of a single type, such as biological networks (Ma, Zhang, Ma, & Ma, 2020; Wu & Ma, 2022). Heterogeneous Information Networks (HINs) are especially advantageous for modeling complex systems because of their diverse node and edge types, and have found applications in various domains, including academic cooperation networks (Wang et al., 2020; Zhang, Luxton-Reilly, Denny and Whalley, 2021), transportation networks (Li, Yu, Shahabi, & Liu, 2018; Zhang et al., 2018), and recommendation systems (Lu, Fang, & Shi, 2020; Zhang, Shi, Zhao, & King, 2019). However, real-world network data often suffer from missing links/relations due to the intricacies and limitations of the data collection process. Using the available information about nodes and edges, we can deduce the underlying principles governing relation formation, thereby enabling us to infer the missing relations. In recent years, a large number of

methods (Hu, Dong, Wang, & Sun, 2020; Lv et al., 2021; Ma, Tan, Xie, Zhong, & Deng, 2022; Nguyen, Liu, & Fang, 2023) have been developed to achieve this goal, while the majority of them primarily concentrate on predicting pairwise relations within the network.

In recent years, numerous studies (Alon, 2007; Milo et al., 2002; Tian et al., 2022; Zhang, Guo et al., 2021) underscored the potential value of higher-order relations in uncovering new insights within complex systems. Many works (Benson, Gleich, & Higham, 2021; Benson, Gleich, & Leskovec, 2016; Lambiotte, Rosvall, & Scholtes, 2019) have emphasized the pivotal role of higher-order graph organization, which forms the core of various graph mining problems in computational biology, chemistry, and other fields. Higher-order relations within networks are typically more intricate, involving more than two nodes, and networks frequently evolve in such higher-order forms (Ceria & Wang, 2023). For instance, consider the academic cooperation network illustrated in Fig. 1(a). In this network, author B publishes a paper A at a conference C, constituting a multi-entity interaction that forms the

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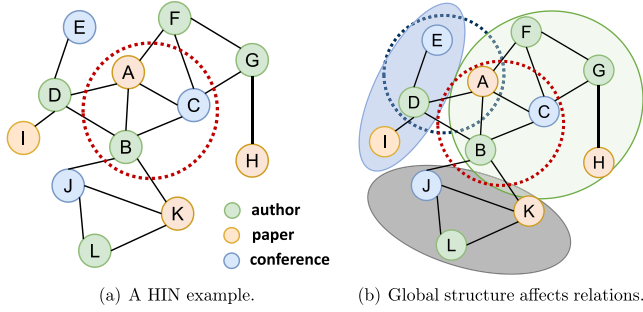


Fig. 1. (a) A HIN example of an academic cooperation network. Different colors represent different node types. The red dashed circle encloses the heterogeneous higher-order relation {A, B, C}. (b) The global structure of the HIN, i.e., community memberships (referring to research fields in academic cooperation networks). Distinct shadow colors represent different communities. Heterogeneous higher-order relation {A, D, E} can be regarded as a counterfactual higher-order relation of {A, B, C}.

higher-order relation {A, B, C}. Such higher-order relations are then incorporated into the known network as local substructures. Therefore, compared with the study of pairwise relations, higher-order relations can provide a deeper understanding of the network structure and capture its essence more effectively. Higher-order interactions are pervasive in various domains. For example, in social networks (Newman, Watts, & Strogatz, 2002), interactions often occur within groups rather than just between two individuals. In biological networks (Navlakha & Kingsford, 2010), many significant phenomena arise from collaborative interactions among multiple proteins, extending beyond the scope of pairwise interactions between two proteins. These studies and examples collectively show the significance of higher-order relations within networks, and the higher-order relation is an extension of pairwise relation, enabling more comprehensive tasks.

Currently, several approaches are available for predicting higher-order relations. One category of approaches utilizes (e.g., triangles, k -node induced subgraphs) to represent higher-order relations. For instance, SEAM (Besta et al., 2022) extracts enclosing subgraphs for multiple nodes of interest. Then it employs a graph neural network to learn features for these subgraphs and predict higher-order relations. However, as shown in Fig. 2(a), subgraphs may not effectively distinguish some more subtle relations. This is because the processed graph data loses some information about the formation process. To address this, another category employs hyperedges to describe higher-order relations. For example, HIT (Liu, Ma, & Li, 2022) uses hyperedges to model four different higher-order relation patterns and captures proximity through random walks to generate structural features used for prediction. Although hyperedges represent a form of higher-order relations, as shown in Fig. 2(b), they do not reveal the internal patterns among a given set of nodes. Hyperedge-based methods usually pay more attention to the interaction at the event level and ignore the interaction pattern inside the event, so they lack flexibility in predicting the relation among any given set of nodes. While these methods exist (Lahiri & Berger-Wolf, 2007; Nassar, Benson, & Gleich, 2020; Yadati et al., 2020), they primarily focus on predicting higher-order relations within homogeneous networks. Research in the context of HINs is still relatively underdeveloped.

In order to understand the formation of higher-order relations deeply, the influence of global graph structure is a key aspect to be learned. For example, as a significant global graph structural feature, the community membership reflects the tight connections between nodes within the network. Nodes within the same community tend to share similar attributes or functions, which often contribute to the formation of edges in the network. Conversely, if nodes do not belong to the same community, the probability of an edge existing between them decreases. Therefore, from a theoretical perspective, there is a causal relationship between community structures and the existence

of edges. Discovering and analyzing this causal relationship is crucial for accurate relationship prediction. From an empirical perspective, learning and integrating global graph structure information improves the performance of downstream tasks in networks (Hao, Qian, Wang, & Bai, 2023; Li, Jian, Wang, & Chen, 2022). For instance, in CAERec (Hao et al., 2023), the node embeddings are learned not only from the semantic information of nodes but also from the similarity of communities, which enhances the quality of embeddings by incorporating community information. Currently, some work has introduced causal learning into graph structures in homogeneous networks (Wang, Feng et al., 2021; Zhao, Liu, Wang, Yu, & Jiang, 2022) and achieved outstanding results. However, there has been no work verifying the impact of global structures on prediction in heterogeneous networks, nor exploring how the causality introduced by global structures can enhance the performance of downstream tasks.

To effectively predict higher-order relations in Heterogeneous Information Networks (HINs), we encounter the following two challenges: **(1) Capturing and Encoding Proximity within Higher-Order Relations.** Higher-order relations in HINs are more intricate due to their involvement with multiple node types. It is essential to devise an appropriate and efficient strategy to capture the multi-node proximity within these complex relations. **(2) Learning the Causal Relationship between Graph Structure and Higher-Order Relation States.** Previous work has largely overlooked the causal relationship aspect. Consider the example in Fig. 1(b) where author B published paper A at conference C, and they share the same community. The strong connection formed within the same community might obscure the underlying factors of their connection, such as common research topics. Moreover, it is possible that being in the same community is itself influenced by these factors. Hence, we propose the counterfactual question: “If these three nodes were not located in the same community, would they still be connected to each other?” If a graph learning model can learn causality by answering this counterfactual question, it can use this knowledge to enhance prediction accuracy.

To tackle the aforementioned challenges, we develop a Counterfactual Learning model for Higher-Order Relation prediction in HINs, shortly named HINCHOR. For challenge (1), we introduce a higher-order structure encoder to capture the proximity within higher-order relations. This encoder employs biased random walks to trace paths within the network and vectorizes the mutual influence of nodes in the higher-order relations. Subsequently, the higher-order relations can be captured by a sequence model, such as LSTM (Graves & Graves, 2012). For challenge (2), we devise the counterfactual data augmentation. In this context, we assume that treatment is a global graph structure feature, such as community membership. We aim to understand how these treatments affect the connections within higher-order relations. Since we cannot provide a definitive answer to the counterfactual question, we redefine the counterfactual higher-order relation. Given a higher-order relation and its treatment, we identify another higher-order relation that fulfills the following two conditions: (i) it possesses different treatment values, and (ii) it is most similar to the given higher-order relation. This identified higher-order relation is referred to as the counterfactual higher-order relation. During prediction, the decoder simultaneously predicts both the factual and counterfactual higher-order relations. In this way, the generated counterfactual higher-order relations can empower the model to learn causal factors.

The contributions of our work are summarized as follows:

- We introduce a novel model for predicting higher-order relations in HINs, named HINCHOR. HINCHOR excels at efficiently capturing proximity information within higher-order relations, marking an early exploration of this aspect within HINs.
- To the best of our knowledge, this is the first instance of incorporating counterfactual learning into higher-order relation prediction. We design a counterfactual data augmentation module to

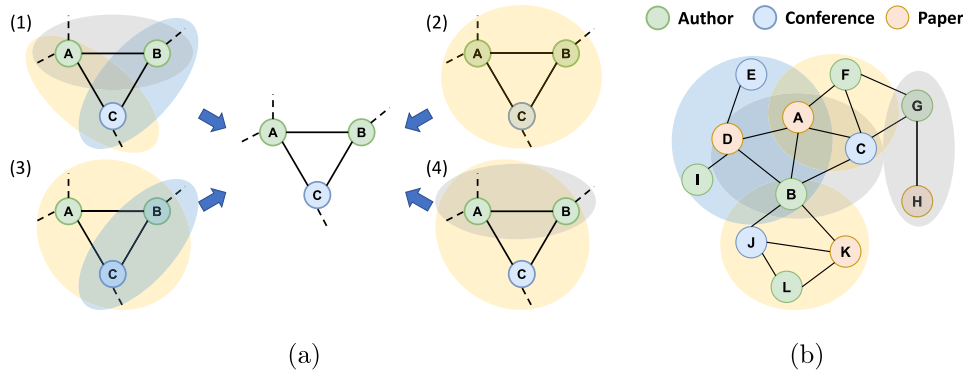


Fig. 2. (a) Same subgraph generated via different hyperedges (represented by shaded areas) (b) Hyperedges are inflexible for describing the relations among any set of nodes.

produce counterfactual higher-order relations. This module serves the purpose of learning the causal relationship between the global graph structure and the states of higher-order relations.

- We construct four distinct benchmark datasets and carry out experiments using them. The results of these experiments indicate that our HINCHOR surpasses the state-of-the-art methods on higher-order relation prediction. Furthermore, our model demonstrates its capacity to measure causal effects.

2. Related work

2.1. Heterogeneous network embedding

One common category of methods for relation prediction on HINs is based on heterogeneous network embedding (HNE) and follows a two-step strategy. After obtaining node or edge representations generated by the HNE method, a simple machine-learning classifier such as Multi-Layer Perceptron (MLP) can be trained to achieve relation prediction. Obviously, the quality of learned representations determines the effectiveness of relation prediction. In recent years, a plethora of heterogeneous network embedding methods have been proposed and can be applied to relation prediction. They can be roughly divided into two categories according to their objectives (Yang, Xiao, Zhang, Sun, & Han, 2020): proximity-preserving methods (Dong, Chawla, & Swami, 2017; Yu et al., 2022; Zhang, Yang, & Xu, 2022; Zhao, Wang, Shi, Liu, & Ye, 2020) and message-passing methods (Fu, Zhang, Meng, & King, 2020; Hu et al., 2020; Li, Huang, Wang, & Ye, 2021; Yang, Li, Zhang, Tang, & Chen, 2021). Proximity-preserving methods capture network topological information by preserving different types of proximity between nodes. For example, TranS (Zhang et al., 2022), a direct relation learning-based method, introduces synthetic relation representations to accommodate complex scenarios where the same entity pair has different relations. Metapath2vec (Dong et al., 2017), which preserves indirect proximity via the random walk strategy, models the semantic context of generated node paths using a language model. Recent method NSHE (Zhao et al., 2020) not only preserves the pairwise proximity but also retains the network schema structure. Message-passing methods aim to learn the target node embedding by aggregating the information of the neighborhood. For instance, StarGAT (Li et al., 2021) proposes a neural network framework to encode nodes with a three-layer hierarchical attention mechanism including node-level, link-level, and motif-level attention. PHGNN (Yang et al., 2021) designs content-level and type-level learning focusing on the importance of edges and nodes respectively, and then hierarchically aggregates neighborhood information. However, these methods are designed to capture pairwise proximity, and when applied to higher-order relation inference involving multiple nodes, they often yield suboptimal results.

In addition, some studies (Dareddy, Das, & Yang, 2019; Ling, Yang, & Zhao, 2023; Yang, Sun, Liu, & Tu, 2017) have leveraged higher-order structures as prior knowledge to enhance the representation

in downstream tasks, achieving good performance in node-level and graph-level tasks. For example, motif2vec (Dareddy et al., 2019) and HGEN (Ling et al., 2023) incorporate motifs as guidance to generate enhanced node-level representations for various downstream tasks. Some methods leverage meta-paths (Guan et al., 2023; Xu et al., 2023) or meta-graphs (Chung & Whang, 2021; Zhang, Fang, Liu, Wu, & Zhang, 2020). For example, HMSG (Guan et al., 2023) learns graph representations from multiple homogeneous and heterogeneous subgraphs generated by various meta-paths. Mg2vec (Zhang et al., 2020) maps meta-graphs into a common node space, guiding node representation learning through first- and second-order embeddings to capture both individual node preferences and their relationships. However, they do not study higher-order relations themselves so that the internal structures within them are not explored. In contrast, our research targets higher-order relations directly, aiming to understand and predict their formation without relying on prior knowledge. The higher-order proximity in Yang et al. (2017) refers to the proximity between two nodes within a multi-hop neighborhood, whereas in our context, it pertains to the proximity within higher-order relations (i.e., among a group of nodes). Therefore, although similar vocabulary is used, the essential challenges and goals of these methods are different from those in this article.

2.2. Higher-order relation prediction

Recent researches have emphasized the importance of higher-order relations (Benson, Abebe, Schaub, Jadbabaie, & Kleinberg, 2018; Benson et al., 2016; Lambiotte et al., 2019), garnering growing attention in the field of network analysis. Some works have focused on predicting higher-order relations, broadly categorized into subgraph prediction (Besta et al., 2022; Lahiri & Berger-Wolf, 2007; Nassar et al., 2020) and hyperedge prediction (Liu et al., 2022; Yadati et al., 2020). On subgraph prediction, the model SEAM (Besta et al., 2022) is an extension of the link prediction method SEAL (Zhang & Chen, 2018). SEAM takes the graph learning model DGCNN (Phan, Le Nguyen, Nguyen, & Bui, 2018) as its backbone, and identifies factors affecting higher-order relation prediction by introducing the concept of “deal breaker edges”. In the work of TRPR (Nassar et al., 2020), a new triangle prediction task is introduced, and traditional link prediction methods are adapted and extended to address this task. Some algorithms (Lahiri & Berger-Wolf, 2007) are also proposed to predict the partial structure of the temporal network. On hyperedge prediction, NHP (Yadati et al., 2020) extends graph neural networks to hypergraphs and proposes a new scoring function that assigns a higher ranking to existing hyperedges compared to non-existent vertex sets. HIT (Liu et al., 2022) is designed to predict higher-order patterns within temporal hypergraphs. It predefines four distinct higher-order relations and characterizes higher-order effects by distance encoding in an inductive learning manner. However, it is worth noting that all of the above focus on the higher-order relations within homogeneous networks, which may have limited expression

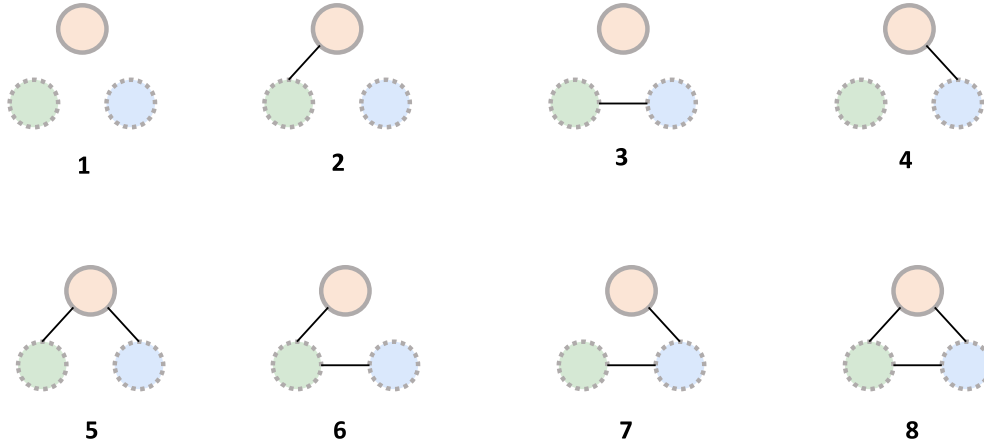


Fig. 3. All the 3-node higher-order relation patterns of the example HIN in Fig. 1. There are a total of 8 3-node patterns. Nodes with solid borders represent the central node type of the HIN, while those with dashed borders denote non-central node types.

capability. In this paper, we aim to address the challenge of higher-order relation prediction on HINs, which represents an existing gap in the research landscape.

2.3. Counterfactual learning

The concept of counterfactual derives from the field of causal inference (Yao et al., 2021), which infers, based on known context, how the outcome would be affected if treatment had been different in the past. Consequently, counterfactual problems generally involve three key factors: context, treatment, and outcome. Counterfactual learning has attracted more and more attention to graph learning and its downstream tasks (Guo, Xiao, Aggarwal, Liu and Wang, 2023), such as fairness learning (Agarwal, Lakkaraju, & Zitnik, 2021; Mehrabi, Morstatter, Saxena, Lerman, & Galstyan, 2021), explanation generation (Abrate & Bonchi, 2021; Lucic, Ter Hoeve, Tolomei, De Rijke, & Silvestri, 2022), link/relation predictions (Chang, Cai, & Li, 2023; Zhao et al., 2022), and recommendation systems (Chen et al., 2022; Liu, Gao, Jin, & Li, 2021). For example, CFLP (Zhao et al., 2022) explores link prediction on homogeneous networks. It aims to uncover the likelihood of two individuals with similar interests becoming friends via the generated counterfactual links. Similarly, KGCF (Chang et al., 2023), a method for knowledge graph completion, instantiates a causal model on the knowledge graph by augmenting the learning from both the observed and counterfactual relations. Since counterfactual instances cannot be directly observed, the crux of these methods lies in how to obtain counterfactuals. In this work, we present the first attempt at generating counterfactual higher-order relations.

3. Preliminary

In this section, we formally define the significant concepts related to the content of this paper. The key notations used in this paper are summarized in Table 1.

Definition 3.1 (Heterogeneous Information Network). A Heterogeneous Information Network (HIN) is formally depicted as a graph $G = (\mathcal{V}, \mathcal{E}, \mathcal{A}, \mathcal{R}, \phi, \psi)$. $\mathcal{V}$ denotes the node set that satisfies a node type mapping function $\phi : \mathcal{V} \rightarrow \mathcal{A}$. $\mathcal{E}$ denotes the edge set which satisfies an edge type mapping function $\psi : \mathcal{E} \rightarrow \mathcal{R}$, and $|\mathcal{A}| + |\mathcal{R}| > 2$.

Definition 3.2 (Heterogeneous Higher-Order Relation). A higher-order relation of the given HIN G is defined as a subgraph $s = (\mathcal{V}_s, \mathcal{E}_s, \mathcal{A}_s, \mathcal{R}_s)$, where $\mathcal{V}_s \in \mathcal{V}$ and $|\mathcal{V}_s| > 2$. It satisfies the injective map $\gamma : \mathcal{V}_s \rightarrow \mathcal{V}$, and $(i, j) \in \mathcal{E}_s$ iff $(\gamma(i), \gamma(j)) \in \mathcal{E}$. Type sets $\mathcal{A}_s$ and $\mathcal{R}_s$ are obtained through the mapping of sets $\mathcal{V}_s$ and $\mathcal{E}_s$ via the function ϕ and ψ , respectively. Note that $\mathcal{V}_s$ contains different types of nodes.

Definition 3.3 (Heterogeneous Higher-Order Relation Pattern). For each heterogeneous higher-order relation s defined in Definition 3.2, there is a corresponding pattern represented as a relation-summarizing graph $P_s = (\mathcal{A}_s, \mathcal{R}_s)$. In $\mathcal{A}_s$, we denote the most central element that connects different node types of the HIN as **central node type** (e.g., the papers in an academic cooperation network), while others are **non-central node types** (e.g., the authors and conferences).

As shown in Fig. 3, here we take the heterogeneous higher-order relation composed of three nodes as an example, where different colors represent different types of nodes. The 3-node pattern set $\mathcal{P}^3$ contains 8 distinct patterns formed by different combinations of edges. In other words, for each 3-node heterogeneous higher-order relation s , $P_s \in \mathcal{P}^3$.

Without loss of generality, in this paper, we focus on 3-node heterogeneous higher-order relations in which all node types are different. The heterogeneous higher-order relations in the following refer to that meaning. On the one hand, the 3-node relations extend upon the 2-node relations, involving multiple edges and covering various types of edges across different node types. On the other hand, in our 3-node relation prediction task, multiple node types and edges are involved, leading to predictions for various types of edges. This issue is consistent with the challenges encountered in predicting other higher-order relations. Thus, the method we adopted in this setting can be easily extended to more complicated higher-order relations.

Definition 3.4 (Causal Effect). The causal effect, also known as the treatment effect, is used to evaluate the difference in outcomes between intervention and non-intervention scenarios, specifically measuring the impact of the treatment variable on the outcome variable. Let W ($W \in \{0, 1\}$) denote the binary treatment, where $W = 1$ represents the treated group and $W = 0$ is the control group. The ATE (Average Treatment Effect) is defined as $\mathbb{E}[Y(W = 1) - Y(W = 0)]$, where $Y(W = 1)$ and $Y(W = 0)$ are the potential treated and control outcome of the whole population respectively.

Definition 3.5 (Prediction Task). Given the observed heterogeneous information network G (the network may be incomplete) and its higher-order relation set $\mathcal{S}$, our goal is to find a function f , for each $s \in \mathcal{S}$, making $f(s) = P_s$.

4. Methodology

Table 1
Mainly used notations.

Notation	Description
$G = (\mathcal{V}, \mathcal{E}, \mathcal{A}, \mathcal{R}, \phi, \psi)$	The input HIN
$s = (\mathcal{V}_s, \mathcal{E}_s, \mathcal{A}_s, \mathcal{R}_s)$	The higher-order relation s of G
P_s, T_s	The pattern and treatment of the s
$S, \mathcal{P}$	The set of the higher-order relations and patterns
s_c	The counterfactual higher-order relation s_c of s
P_{s_c}, T_{s_c}	The pattern and treatment of the s_c
$\mathcal{V}_{s_c}$	The nodes set of s_c
$\{v_i^c, v_c^c, v_j^c\}$	v_c^c is central node while others are non-central nodes
$h(\cdot, \cdot), l(\cdot, \cdot)$	The distance functions of node sets and vectors
$\tilde{\mathbf{X}}$	The node embeddings of G for distance calculation
α_{tw}	The unnormalized walk probability of the next node v_w from node v_t
d_{tw}	The distance between t and w in G
b, q	The parameters control random walk bias towards BFS or DFS.
$\mathcal{W} = \{\mathcal{W}_i v_i \in \mathcal{V}_s\}$	The walk set of the s
$m(v_u; \mathcal{W}_i), m(v_u; \mathcal{W})$	The context encoding vector of node v_u based on $\mathcal{W}_i$ and $\mathcal{W}$
$g(v_u; \mathcal{W})$	The heterogeneous context encoding vector of node v_u
p_W^i	The vector of walk W starting with the source node v_i
$\mathbf{x}_i$	The learned embedding of node v_i
$\hat{P}_s, \hat{P}_{s_c}$	The predicted patterns of s and s_c

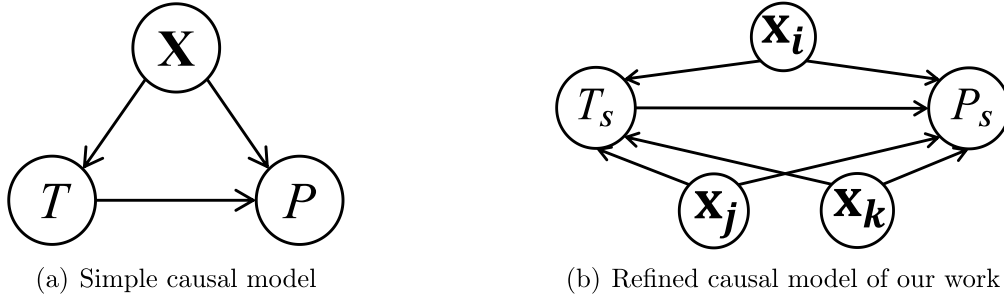


Fig. 4. (a) Simple causal structure where context $\mathbf{X}$ confounds the effect of treatment T on the outcome P . (b) Given the higher-order relation $s = \{i, j, k\}$, the refined causal model on higher-order relation prediction based on node embeddings $\{\mathbf{x}_i, \mathbf{x}_j, \mathbf{x}_k\}$. Achieve better predictions of higher-order relation by evaluating the causal effect between T_s (global structure) and P_s (predicted outcome).

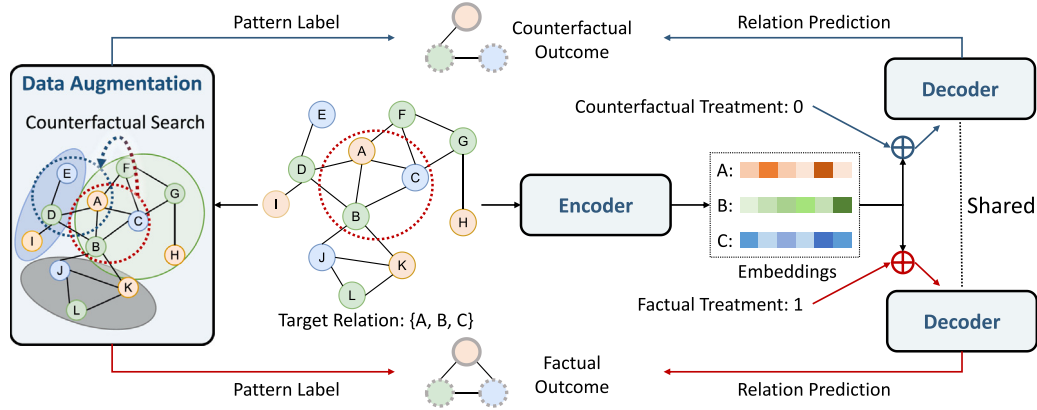


Fig. 5. The overall architecture of our proposed HINCHOR. For higher-order relation $\{A, B, C\}$, $\{A, D, E\}$ is the counterfactual higher-order relation obtained through counterfactual search. After processing through a higher-order structure encoder, both factual and counterfactual higher-order relations are fed into a decoder to predict their patterns under different treatment variables.

4.1. Overview

The simple structural causal model on higher-order relation prediction can be summarized as shown in Fig. 4(a). The context $\mathbf{X}$ (i.e., the HIN) confounds the effect of treatment T (i.e., the global structure) on the outcome P (i.e., the higher-order relations). Thus, to achieve better prediction, the proposed predictor needs to mitigate the confounding effects. In other words, we should endow the predictor with the ability to study the causal relationship between global structure and higher-order relations.

In this work, we refine the above structural causal model as illustrated in Fig. 4(b). Then we introduce our method, HINCHOR, following the refined causal model with counterfactual causal inference. The overall framework of HINCHOR is shown in Fig. 5. The target higher-order relation s , which consists of nodes v_i, v_j and v_k , undergoes processing through three modules: (a) Counterfactual Data Augmentation: This module, taking the global structural feature as treatment, generates corresponding counterfactual relation with treatment states opposite to the given s ; (b) Higher-Order Structure Encoder: This component captures the multi-node proximity within higher-order

relations, transforming the context information into representations of nodes v_i , v_j and v_k ; (c) Higher-Order Relation Pattern Decoder: It is employed to give both the factual and counterfactual outcomes, i.e., the higher-order relation patterns, for the given s based on fused node representations and treatments. We provide further details for each module below.

4.2. Counterfactual data augmentation

To conduct counterfactual causal inference, it is essential to determine the outcomes when treatments differ from the observed ones (Yao et al., 2021). Regarding treatments, it is possible to study any binary property of the given higher-order relation s . To maintain generality, we select one of the most significant global structural features, community membership, as the treatment of interest. Formally, consider a function $c(\cdot)$ as a community mapping function (e.g., SBM Karrer & Newman, 2011), where its input is nodes, and its output is the community label of each node. We assign the treatment values to indicate whether nodes within the higher-order relation belong to the same community:

$$T_s = \begin{cases} 1 & \text{if } c(v_i) = c(v_j) = c(v_k) \wedge \mathcal{V}_s = \{v_i, v_j, v_k\}, \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

However, in the real world, we typically only have access to **observational factual data**. Therefore, the key challenge lies in how to infer those counterfactual outcomes. Inspired by recent works (Chang et al., 2023; Zhao et al., 2022), we adopt a data augmentation strategy of searching for other higher-order relation instances having a similar context but different treatment to serve as counterfactual substitutes. Based on this idea, for each higher-order relation s , we use the pattern of its nearest higher-order relation with the opposite treatment as the alternative counterfactual outcome. Formally, the counterfactual higher-order relation s_c of the higher-order relation s :

$$\mathcal{V}_{s_c} = \arg \min \left\{ h(\mathcal{V}_s, \mathcal{V}_{s_c}) \mid T_{s_c} = 1 - T_s \right\}, \quad (2)$$

where $h(\cdot, \cdot)$ is a measure of the distance between a pair of node sets (a pair of higher-order relations).

Solving Eq. (2) is a computationally expensive process that is nearly impossible to scale. Iterating through all combinations of 3 nodes and calculating the shortest distance to find the nearest higher-order relation has a time complexity of $O(|\mathcal{V}|^6)$. Thus, we propose a node-embedding-based approach. We obtain a set of temporary node embeddings $\tilde{\mathbf{X}} \in \mathbb{R}^{|\mathcal{V}| \times F}$ through a classic HNE model (e.g., Metapath2vec Dong et al., 2017). Subsequently, we apply the distance function $l(\cdot, \cdot)$ (e.g., Euclidean distance) on the embedding space of $\tilde{\mathbf{X}}$, which is significantly more efficient than calculating the distance directly on the HIN. Furthermore, to simplify the search process and ensure that the discovered counterfactual higher-order relations are more meaningful, we keep the central node fixed and only search for the nearest neighbors of those non-central nodes. As for the example in Fig. 1(b), we fix the paper node and find the nearest neighbors of the author and conference nodes. For a higher-order relation s , the found counterfactual s_c whose node set $\mathcal{V}_{s_c} = \{v_i^c, v_j^c, v_k^c\}$ satisfies:

$$\begin{aligned} (v_i^c, v_j^c, v_k^c) = \arg \min_{v_i^c, v_j^c, v_k^c \in \mathcal{V}} \left\{ l(\tilde{\mathbf{x}}_i, \tilde{\mathbf{x}}_i^c) + l(\tilde{\mathbf{x}}_j, \tilde{\mathbf{x}}_j^c) \mid \right. \\ \left. T_{s_c} = 1 - T_s, l(\tilde{\mathbf{x}}_i, \tilde{\mathbf{x}}_i^c) + l(\tilde{\mathbf{x}}_j, \tilde{\mathbf{x}}_j^c) < \gamma \right\}, \end{aligned} \quad (3)$$

where v_i^c is the central node of the s_c and it is equal to the central node v_i of s . v_j^c and v_k^c are the non-central nodes. $\tilde{\mathbf{x}}_i^c$ (or $\tilde{\mathbf{x}}_j^c$) is the embedding vector of the v_i^c (or v_j^c). T_{s_c} is the treatment of s_c . γ is a hyperparameter that defines the maximum distance at which two nodes are considered similar. If no counterfactual higher-order relation meeting the specified conditions is found for s , we will assign it the same treatment and pattern as the factual higher-order relation:

$$(T_{s_c}, P_{s_c}) = \begin{cases} 1 - T_s, P_{s_c} & \text{if } \exists s_c \text{ (Eq. (3))}, \\ T_s, P_s & \text{otherwise,} \end{cases} \quad (4)$$

where P_s is the factual outcome (pattern) of the higher-order relation s and P_{s_c} is oppositely the counterfactual outcome.

After the above search, for the vast majority of higher-order relations, their counterfactual outcomes can be obtained. We utilize the augmented data to enhance the training process. Therefore, the higher-order relation predictor takes s as input but both P_s and P_{s_c} as the labels.

4.3. Higher-order structure encoder

Although many heterogeneous graph neural networks (HGNNs) have been proposed to encode HIN, they typically capture the proximity between two nodes (Rossi, Jin et al., 2020), lacking the capability to capture interaction patterns among groups of nodes. Furthermore, HGNNs may suffer from over-smoothing, making it hard to infer their specific internal structure, as the embeddings of nodes become similar to those of their neighboring nodes. Hence, there is a need to design a particular encoder to learn complex interaction patterns among groups of nodes.

In the context of the HIN, the most intuitive and precise way to describe higher-order structure is through heterogeneous motifs (Rossi, Ahmed et al., 2020). However, enumerating and encoding heterogeneous motifs comes with significant computational costs. Therefore, we extend the anonymous walk strategy to heterogeneous higher-order relations for approximation (Guo, Jiao et al., 2024; Wang, Chang, Liu, Leskovec and Li, 2021). As shown in Fig. 6, multiple distinct random walks can be mapped to the same anonymous path. This allows us to ignore their specific identities and focus on their connectivity patterns. These anonymous paths can be seen as different subgraphs, which are considered structural and functional blocks of the network. For instance, motif (1) provides detailed information about the publication, including the authors and conference details, and motif (2) represents multiple authors who have published papers at the same conference. Utilizing these higher-order blocks helps us understand the essence of networks.

Based on the approximation, we design a higher-order structure encoder to capture and encode the multi-node proximity with the higher-order relations. The overview of the whole encoding process is given in Fig. 7. To confine the scope of traversed nodes in the vicinity of the target higher-order relation, we bias the random walk procedure (Grover & Leskovec, 2016) with the parameters b, q . For a node v_u in the random walk, suppose that its previous node is v_i , then the unnormalized probability of the next node v_w is:

$$\alpha_{tw} = \begin{cases} \frac{1}{b} & \text{if } d_{tw} = 0, \\ 1 & \text{if } d_{tw} = 1, \\ \frac{1}{q} & \text{if } d_{tw} = 2, \end{cases} \quad (5)$$

where d_{tw} is the distance between t and w in the HIN. With low b and high q , we can keep the walk close to the source node. For each node v_i in the higher-order relation s , we use it as the source node to sample a number of biased walks with length n . Therefore, the neighborhood information of each node v_i can be captured by its walk set $\mathcal{W}_i$, where we set $|\mathcal{W}_i| = N$. The context structure information of the higher-order relation s can be represented by the three corresponding walk sets $\mathcal{W} = \{\mathcal{W}_i \mid v_i \in \mathcal{V}_s\}$.

In order to integrate the higher-order structure information into the representations, we encode the nodes of higher-order relation based on its context $\mathcal{W}$. Each node v_u that occurs in context $\mathcal{W}$ at least once is either a direct or indirect neighbor of the source nodes, and this allows us to capture the inter-walk and intra-walk dependencies among the nodes within the target higher-order relation. For each walk set $\mathcal{W}_i \in \mathcal{W}$, we encode each context node v_u appears in $\mathcal{W}$ as a $(n+1)$ -dimension vector $m(v_u; \mathcal{W}_i)$. The a th ($a = 0, \dots, n$) entry is:

$$m(v_u; \mathcal{W}_i)[a] = \left| \left\{ W \mid W \in \mathcal{W}_i, v_u \text{ occurs at } W[a] \right\} \right|. \quad (6)$$

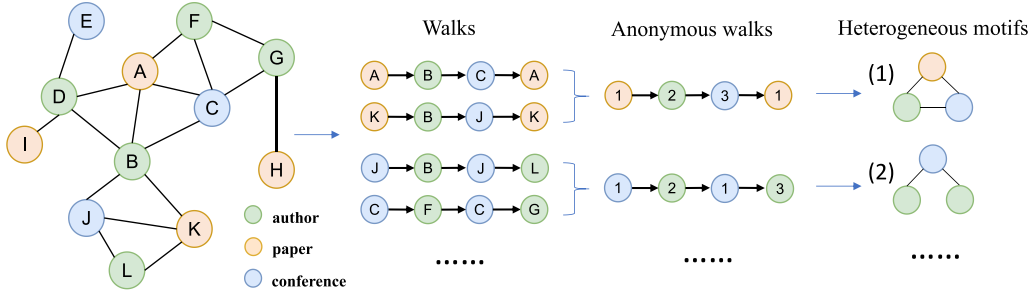


Fig. 6. Anonymous walks approximate the heterogeneous motifs. Different walks may have the same anonymous walks, denoting the same heterogeneous motifs.

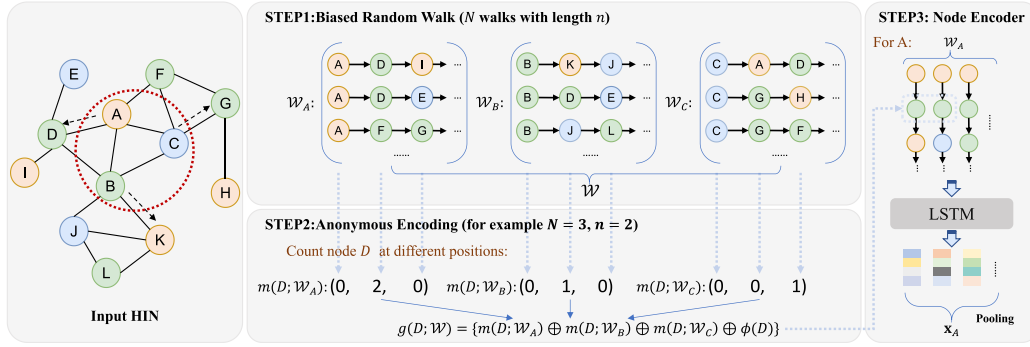


Fig. 7. The higher-order structure encoder of our proposed HINCHOR. First, perform a biased random walk for each node in the higher-order relation $\{A, B, C\}$ to obtain its path set W , thereby modeling its context. Next, anonymize the nodes as $g(\cdot; W)$ to learn higher-order proximities. Finally, use LSTM and pooling methods to encode the context information into the node representations.

In essence, $m(v_u; \mathcal{W}_i)$ is calculated based on the frequency of node v_u appearing at a specific position in $\mathcal{W}_i$. Similarly, for each source node v_i , i.e., the node within the target higher-order relation s , we still represent it in the above way. But we set the first dimension of $m(v_i; \cdot)$ to 0 on all walk sets, so as to avoid the counting bias caused by the walks starting with source nodes.

On the whole context $\mathcal{W}$, we concatenate $m(v_u; \cdot)$ on all walk sets $\mathcal{W}_i, \mathcal{W}_j, \mathcal{W}_k \in \mathcal{W}$ as the anonymous encoding for a node v_u :

$$m(v_u; \mathcal{W}) = m(v_u; \mathcal{W}_i) \oplus m(v_u; \mathcal{W}_j) \oplus m(v_u; \mathcal{W}_k), \quad (7)$$

$\oplus$ is the concatenation operator. Due to the heterogeneity of nodes, we also take into account the type of nodes. Therefore, the heterogeneous context nodes are anonymously encoded as:

$$g(v_u; \mathcal{W}) = m(v_u; \mathcal{W}) \oplus \phi(v_u). \quad (8)$$

After obtaining the anonymous encoded vectors of all nodes in the context $\mathcal{W}$, we represent each walk $W \in \mathcal{W}_i$ starting with the source node v_i ($v_i \in \mathcal{V}_s$) as:

$$p_W^i = \{g(W[a]; \mathcal{W}) | a = 0, 1, \dots, n\}. \quad (9)$$

Through the above anonymous encoding, the encoded sequences effectively capture the positional relationships and interaction frequencies between each context node and the target node, serving as an approximate representation of the high-order structure in the target node's neighborhood. Then, we employ neural networks to encode all the walks starting with the source node v_i and apply a set pooling method AGG to calculate its final representation:

$$\mathbf{x}_i = \text{AGG}(\{\text{Enc}(p_W^i)\}_{W \in \mathcal{W}_i}), \quad (10)$$

p_W^i is a long sequence containing numerous zero elements, where the positions and values of non-zero elements are crucial. Therefore, the encoder Enc must effectively retain the essential ordered information within the sequence, capturing both implicit and long-term dependencies. To achieve this, we utilize LSTM, a sequential neural model capable of learning both local and global patterns in the data, as the encoder Enc. We instantiate AGG as mean pooling.

4.4. Higher-order relation pattern decoder

The decoder takes the representations of nodes v_i, v_j and v_k within the higher-order relation s as input, and discerns the factual and counterfactual patterns of s . For this classification task, we adopt a simple MLP as the decoder. To control the outcome state, that is, to determine whether the outcome is factual or counterfactual, we inject the treatment into the decoder as well:

$$\hat{P}_s = \text{MLP}(\mathbf{x}_i \oplus \mathbf{x}_j \oplus \mathbf{x}_k \oplus T_s), \quad (11)$$

$$\hat{P}_{s_c} = \text{MLP}(\mathbf{x}_i \oplus \mathbf{x}_j \oplus \mathbf{x}_k \oplus T_{s_c}), \quad (12)$$

where $\hat{P}_s$ and $\hat{P}_{s_c}$ are the predicted factual and counterfactual outcomes, respectively. As both factual and counterfactual outcomes can be obtained via this predictor, we can further estimate the causal effect between the treatments and outcomes, i.e., the global structure and higher-order relations.

4.5. Optimization objective

Both factual and counterfactual higher-order relation prediction tasks are multi-class classification, we use the cross entropy loss for optimization:

$$Loss_f = - \sum_{s \in S} \sum_{k=1}^K P_s^k \log(\hat{P}_s^k), \quad (13)$$

$$Loss_{cf} = - \sum_{s \in S} \sum_{k=1}^K P_{s_c}^k \log(\hat{P}_{s_c}^k), \quad (14)$$

where K is the number of classes. Therefore, the overall training loss of our model is:

$$Loss = Loss_f + \beta Loss_{cf}, \quad (15)$$

where β is the hyperparameter regulating the trade-off between factual and counterfactual learning.

Algorithm 1 The learning procedure of HINCHOR

Require: The HIN G , the initial embeddings $\tilde{X}$, the higher-order relations set S and corresponding patterns $\{P_s\}_{s \in S}$; the hyperparameters b, q, n, N and β ; the epoch number E .

Ensure: The model parameters Θ of HINCHOR

- 1: Get global features from G and factual treatment $\{T_s\}_{s \in S}$ via Eq. (1);
 - 2: Search for the counterfactuals $\{s_c\}_{s \in S}$ via Eq. (3) based on $\tilde{X}$, and get corresponding counterfactual patterns $\{P_{s_c}\}_{s \in S}$;
 - 3: Generate biased walk set $\mathcal{W}$ by random walk via Eq. (5);
 - 4: Calculate anonymous encodings on $\mathcal{W}$ via Eqs. (6)–(8);
 - 5: Get walk feature p_W via Eq. (9);
 - 6: **for** $epoch = 1$ to E **do**
 - 7: **for** $s \in S$ **do**
 - 8: Encode p_W^s with LSTM as $\{x_i\}_{i \in V_i}$ via Eq. (10);
 - 9: Decode factual pattern of s via Eq. (11);
 - 10: Decode counterfactual pattern of s via Eq. (12);
 - 11: Update Θ via Eqs. (13)–(15);
 - 12: **end for**
 - 13: **end for**
 - 14: **return** Model parameters Θ ;
-

4.6. Complexity analysis

We analyze the complexity of HINCHOR module by module. (1) Searching for counterfactual higher-order relations: The time complexity is $O(|S|)$ because finding a counterfactual for each higher-order relation costs constant time with a small γ . (2) Biased random walk for encoding: The preprocessing time for calculating the AliasTable is $O(|V|)$, and the sampling time for one node is $O(1)$. Therefore, the time complexity for sampling N walks with length n is $O(nN|S|)$. LSTM encodes a total of $3N|S|$ walks, costing time of $O(nNH|S|(H+n))$, where H is the dimension of the hidden states. (3) Decoder (MLP): The time complexity of the decoder is $O(H'H|S|)$, where H' is the number of hidden layer units.

Remark. The pseudocode of the whole learning procedure is shown in Algorithm 1. In line 1, a community detection algorithm is utilized to obtain community membership for each node, which provides the treatment for factual higher-order relations. In line 2, based on embedding $\tilde{X}$, HINCHOR searches for the counterfactual higher-order relations and gets corresponding counterfactual patterns. In lines 3–5, the model performs random walks to acquire context information, generates node features based on these walk sets, and constructs walk features. In lines 6–8, for each higher-order relation, HINCHOR encodes the contexts and aggregates them. In lines 9–10, the decoder is used to predict the patterns of factual and counterfactual higher-order relations. In line 11, loss is computed for updating the parameters. After training, we can use the model to predict higher-order relations.

Note that we used the 3-node heterogeneous relationship above as an example for easier description and understanding. Our encoder generates walks for nodes of higher-order relation, without limiting to just three nodes. In augmentation, multiple central nodes can be fixed to ensure semantic overlap and search efficiency. Increasing the decoder's dimensions can accommodate more nodes. Thus, our method can be easily extended to higher-order relations with more nodes.

5. Experiments

5.1. Experimental setup

The task of higher-order relation prediction on HINs, explored in this work, is a new task that has not been practiced in previous research. Therefore, we also put effort into establishing benchmarks for this task.

Datasets. We build a benchmark on four real-world HIN datasets including **DBLP**, **DouBan**, **Aminer** and **IMDB**. The statistics are summarized in Table 2. The details of these datasets are as follows:

- (1) **DBLP**¹: This is a computer science bibliography website. We extract a subset of DBLP which contains 14 376 papers (P), 14 475 authors (A), 20 conferences (C), and 8920 terms (T). The types of target nodes that we use to construct 3-node higher-order relations are author, paper and conference.
- (2) **DouBan**²: We extract a subset of DouBan Book which contains 3000 books (B), 2141 authors (A), 615 publishers (P). These three types are used to construct 3-node higher-order relations.
- (3) **Aminer**³: This is a academic social network. We extract a subset of Aminer which contains 5000 papers (P), 13 143 authors (A), and 280 conferences (C). These three types are used to construct 3-node higher-order relations.
- (4) **IMDB**⁴: This is a movie review database. We construct the heterogeneous network with 943 users (U), 1682 movies (M), 5 ratings (R), and 19 genres (G). The types of target nodes that we use to construct 3-node higher-order relations are user, movie and genre.

In networks, communities represent subsets of nodes that are tightly connected within themselves while having relatively sparse connections with other subsets. In citation networks such as DBLP and Aminer, community membership refers to research fields; in IMDB, it refers to movie genres; and in DouBan, it refers to book categories.

We randomly remove 20% of the edges in the original HIN, and then extract a number of heterogeneous higher-order relations (the number is shown in Table 2). Due to the characteristics of the network, pattern 5 shown in Fig. 3 does not exist. Take DBLP as the example, if an author and a paper are connected, the paper and the conference where it is published are connected. It means that the author has published the paper at the conference, so we connect the author and the conference. For the rest patterns, their quantities are balanced in the dataset. We divide the dataset into training, validation, and test sets in a ratio of 3:1:1.

Baselines. No existing methods are suitable for the higher-order relation prediction task proposed in this paper. Therefore, we extend two categories of baseline methods to this new task for comparison. The first category comprises embedding-based models, such as DeepWalk (Perozzi, Al-Rfou, & Skiena, 2014), Metapath2vec (Dong et al., 2017), RHINE (Lu, Shi, Hu, & Liu, 2019), HIN2vec (Fu, Lee, & Lei, 2017), GATNE (Cen et al., 2019), NSHE (Zhao et al., 2020) and

¹ <https://github.com/cynricfu/MAGNN>.

² <https://github.com/librahu/HIN-Datasets-for-Recommendation-and-Network-Embedding>.

³ <https://www.aminer.cn/citation>.

⁴ <https://grouplens.org/datasets/movielens/100k/>.

Table 2
Statistics of datasets.

Dataset	#Nodes	#Edges	#Higher-order relations
DBLP	A: 14,475; P: 14,376; C: 20; T: 8920	P-A: 41,794; P-C: 14,376; P-T: 114,624; A-C: 24,495	8750
DouBan	A: 2141; B: 3000; P: 615	A-B: 3000; B-P: 3000; A-P: 2531	8590
Aminer	A: 13,143; P: 3000; C: 280	P-A: 14,331; P-C: 5000; A-A: 19,165; A-C: 13,663	8529
IMDB	U: 943; M: 1682; R: 5; G: 19	U-M: 100,000; U-R: 4423; M-G: 2893; M-R: 6714; U-G: 14,236	14,000

HeGAN (Hu, Fang, & Shi, 2019). For these methods, we concatenate the embeddings of the multiple nodes within higher-order relation as the relation representation, then train a simple MLP classifier to make predictions. The second category includes heterogeneous link prediction models like NOH (Lu et al., 2023), HeteHG-VAE (Fan et al., 2021), HGT (Hu et al., 2020), SLICE (Wang, Agarwal, Ham, Choudhury and Reddy, 2021) and SimpleHGN (Lv et al., 2021). Additionally, we apply GraphGPT (Tang et al., 2024), a recently proposed method that integrates GNN with the popular large language model (LLM), as a link predictor. We perform link prediction on each node pair within the higher-order relation separately and combine all the links as the final predicted pattern. For methods using metapaths, we predefine the metapaths on the four datasets as APCPA, ABPBA, APCPA and UMGMU, respectively. For baselines involving random walks, we sample 200 walks per node with a length of 50, and the window size is 5. For GraphGPT, we utilize the pre-trained parameters provided by its authors and further perform task-specific instruction tuning on our datasets. We feed the type information into the LLM as textual input. In terms of other parameters, we try to follow the settings described in the original paper.

- (1) **DeepWalk** (Perozzi et al., 2014) is an unsupervised graph embedding method primarily designed for homogeneous graphs. It leverages the co-occurrence relationships between nodes captured by random walks to learn embeddings.
- (2) **Metapath2vec** (Dong et al., 2017) is an extension of DeepWalk specifically designed for HINs. It introduces symmetric metapaths that control the random walk sampling process.
- (3) **RHINE** (Lu et al., 2019) divides heterogeneous relations into Affiliation Relations and Interaction Relations. Then it uses different models to handle these two types of relations, aiming to capture both the structural and semantic information of the HIN more effectively.
- (4) **HeGAN** (Hu et al., 2019) is inspired by the generative adversarial network framework. It trains discriminators and generators simultaneously in minimax games to learn node embeddings on HIN.
- (5) **HIN2vec** (Fu et al., 2017) learns node embeddings by predicting whether there is a specific relation between two given nodes in HIN. It generates training data based on random walks and negative sampling.
- (6) **GATNE** (Cen et al., 2019) is designed to address the problem of learning node embeddings in heterogeneous multiplex networks. GATNE generates node embeddings by combining base embeddings and edge embeddings.
- (7) **NSHE** (Zhao et al., 2020) proposes a network schema sampling method to generate subgraphs and constructs a multi-task learning task to preserve pairwise structure information and higher-order structures (i.e., network schema) in the HIN.
- (8) **NOH** (Lu et al., 2023) uses two message-passing modules to capture hyperedge-based higher-order semantics and lower-order pairwise relations. The latter enhances the learning of local information by capturing neighborhood overlap.

- (9) **HeteHG-VAE** (Fan et al., 2021) constructs a heterogeneous hypergraph to capture higher-order semantics. It introduces a unsupervised Bayesian deep generative framework to generate node embeddings.
- (10) **HGT** (Hu et al., 2020) learns node embeddings and performs link prediction on HIN. It applies the Transformer model and uses meta-relations to parameterize the weight matrix so that different types of nodes and edges can maintain their specific representation space.
- (11) **SLICE** (Wang, Agarwal et al., 2021) is a self-supervised learning approach on HIN. It pre-trains the model by introducing higher-order semantic associations and masking nodes to learn contextual node embeddings and then fine-tunes them for the link prediction task.
- (12) **SimpleHGN** (Lv et al., 2021) takes the graph attention network as the backbone and improves it in three components: learnable edge-type embedding, residual connections, and L2 normalization. It can be used for link prediction on HINs.
- (13) **GraphGPT** (Tang et al., 2024) is a LLM designed for homogeneous graphs. It bridges textual information and graph structures through a text-graph grounding stage and aligns graph representations with textual instructions via the dual-stage graph instruction tuning. This enables it to handle link prediction tasks in a question–answering format.

Our Method. For HINCHOR, the length n of random walks is 10 and the number N of walks per node is 300. The embedding dimension is set as 128. The training epoch is 200 and the learning rate is dynamically adjusted by the Adam optimizer (Kingma & Ba, 2015) during the learning process. Hyperparameters b, q, γ, β are set to 1, 2, 0.5, 1, respectively. We apply early stopping with patience 20. For the global structure feature, we use the stochastic block model (SBM) (Karrer & Newman, 2011) by default. Metapath2vec is employed to obtain the temporary representation matrix $\tilde{X}$. The experiments are conducted in an Ubuntu 18.04 environment with an NVIDIA GeForce RTX 3090 (24 GB) GPU, using Python 3.6 and the PyTorch deep learning framework. More details of our model are available at <https://github.com/meteroy/HINCHOR>.

For both the baseline methods and our proposed HINCHOR, we repeat the experiments ten times to calculate the mean $\pm$ standard deviation of the results.

5.2. Prediction results

We first evaluate the performance of HINCHOR on higher-order relation prediction. We use multiple metrics on multi-class classification to provide a more comprehensive evaluation. The results measured by Macro-F1, Micro-F1 and NMI are reported in Table 3. The results measured by Recall, Precision and AUC are reported in Table 4. The best results are in bold.

It can be clearly observed that the proposed HINCHOR significantly outperforms all the other methods on four datasets. It demonstrates that our model can effectively capture higher-order proximity and encode them into meaningful features. In DBLP and DouBan, the best baseline

Table 3

Prediction results (%) measured by Macro-F1, Micro-F1 and NMI.

Method	DBLP			DouBan			Aminer			IMDB		
	Macro-F1	Micro-F1	NMI	Macro-F1	Micro-F1	NMI	Macro-F1	Micro-F1	NMI	Macro-F1	Micro-F1	NMI
DeepWalk	38.0 ± 2.6	44.3 ± 1.1	44.7 ± 5.0	42.4 ± 5.3	46.2 ± 6.1	56.1 ± 7.0	64.0 ± 1.7	68.9 ± 1.6	76.5 ± 1.5	41.7 ± 2.8	47.6 ± 1.7	70.3 ± 3.6
Metapath2vec	34.3 ± 13.6	40.3 ± 11.0	35.2 ± 10.5	48.4 ± 5.3	48.2 ± 3.4	53.6 ± 6.8	43.8 ± 5.0	50.7 ± 3.1	61.4 ± 2.3	41.2 ± 1.6	44.7 ± 1.5	68.5 ± 3.2
RHINE	15.5 ± 8.4	24.8 ± 8.7	22.8 ± 14.2	23.2 ± 16.1	31.7 ± 15.0	29.4 ± 25.5	6.0 ± 3.2	16.0 ± 2.6	4.8 ± 0.7	22.8 ± 8.0	28.9 ± 4.7	50.5 ± 15.5
HeGAN	49.2 ± 1.4	50.2 ± 1.0	45.7 ± 0.7	51.0 ± 7.2	50.2 ± 7.0	42.4 ± 2.7	59.9 ± 1.7	64.4 ± 1.8	69.4 ± 1.8	41.7 ± 2.6	45.3 ± 2.2	70.3 ± 2.5
HIN2vec	15.1 ± 4.6	24.6 ± 3.5	19.9 ± 5.4	14.4 ± 5.7	23.9 ± 5.8	25.9 ± 7.6	34.7 ± 7.5	41.9 ± 5.7	51.0 ± 7.2	26.8 ± 8.8	43.3 ± 9.1	39.2 ± 13.3
GATNE	27.1 ± 9.8	36.3 ± 8.5	40.2 ± 12.6	41.1 ± 16.5	46.8 ± 14.9	52.3 ± 2.1	63.5 ± 1.4	68.9 ± 0.7	75.3 ± 0.6	47.4 ± 11.8	53.6 ± 6.6	59.5 ± 6.7
NSHE	28.6 ± 8.9	38.3 ± 7.4	35.1 ± 9.2	33.1 ± 11.0	40.6 ± 11.5	46.3 ± 13.4	56.6 ± 13.4	62.3 ± 12.2	68.6 ± 12.4	47.2 ± 1.5	53.8 ± 1.1	55.6 ± 1.0
NOH	37.4 ± 2.8	43.6 ± 1.4	46.0 ± 6.5	49.2 ± 3.6	48.2 ± 3.5	40.9 ± 1.7	31.9 ± 2.3	36.0 ± 3.1	45.4 ± 5.3	35.6 ± 17.5	41.5 ± 13.3	32.7 ± 11.1
HeteHG-VAE	46.2 ± 1.1	51.5 ± 1.4	45.6 ± 0.7	47.7 ± 8.6	49.0 ± 6.2	44.4 ± 3.9	28.4 ± 9.7	34.8 ± 8.0	34.6 ± 13.7	51.0 ± 5.0	51.6 ± 4.1	51.3 ± 5.8
HGT	51.9 ± 3.0	54.8 ± 2.3	51.1 ± 2.1	59.4 ± 1.2	62.2 ± 0.7	65.2 ± 1.2	34.6 ± 7.0	42.4 ± 8.1	40.0 ± 7.0	42.6 ± 6.4	50.7 ± 3.9	65.4 ± 2.7
SLICE	11.0 ± 0.7	12.6 ± 0.5	14.9 ± 1.1	11.1 ± 1.2	13.0 ± 1.2	3.9 ± 1.1	10.6 ± 1.0	12.2 ± 0.9	5.4 ± 2.4	10.5 ± 1.1	11.9 ± 1.3	4.9 ± 0.8
SimpleHGN	7.1 ± 0.6	12.4 ± 2.2	32.4 ± 8.3	36.9 ± 5.1	45.9 ± 4.6	32.2 ± 4.1	34.6 ± 7.0	42.4 ± 8.1	40.0 ± 7.0	7.1 ± 0.6	12.4 ± 2.2	32.4 ± 8.3
GraphGPT	3.4 ± 0.0	13.4 ± 0.0	0.0 ± 0.0	3.3 ± 0.0	13.4 ± 0.0	0.0 ± 0.0	3.6 ± 0.0	14.2 ± 0.0	0.0 ± 0.0	4.4 ± 0.0	18.3 ± 0.0	0.0 ± 0.0
HINCHOR w/o CF	89.1 ± 0.8	89.2 ± 0.8	77.2 ± 1.0	87.9 ± 3.0	88.3 ± 2.9	80.4 ± 2.7	88.1 ± 1.2	88.7 ± 1.2	85.5 ± 1.8	80.0 ± 1.1	83.1 ± 1.3	72.4 ± 1.4
HINCHOR	90.0 ± 0.8	90.1 ± 0.8	78.7 ± 1.4	88.5 ± 2.4	88.7 ± 2.3	80.9 ± 2.2	89.0 ± 0.6	89.4 ± 0.6	85.6 ± 3.3	78.2 ± 6.6	83.5 ± 2.9	71.9 ± 2.7

Table 4

Prediction results (%) measured by Recall, Precision, and AUC.

Method	DBLP			DouBan			Aminer			IMDB		
	Recall	Precision	AUC	Recall	Precision	AUC	Recall	Precision	AUC	Recall	Precision	AUC
DeepWalk	44.3 ± 1.1	41.3 ± 4.7	68.3 ± 0.4	45.6 ± 6.4	44.2 ± 6.7	70.3 ± 1.4	69.5 ± 1.6	67.2 ± 1.0	83.4 ± 0.2	50.3 ± 3.7	45.7 ± 2.5	75.3 ± 1.0
Metapath2vec	40.3 ± 11.0	35.9 ± 13.6	71.0 ± 2.1	47.6 ± 3.2	54.9 ± 6.3	66.9 ± 0.5	51.3 ± 3.0	59.0 ± 5.2	80.2 ± 0.4	48.0 ± 1.3	43.3 ± 2.6	74.5 ± 1.0
RHINE	24.8 ± 8.7	18.8 ± 7.4	69.1 ± 0.7	28.8 ± 11.4	19.1 ± 9.2	70.6 ± 2.9	15.5 ± 2.6	7.0 ± 6.1	65.8 ± 2.4	30.3 ± 8.0	25.3 ± 9.8	70.4 ± 2.5
HeGAN	50.2 ± 1.0	50.7 ± 0.7	78.0 ± 0.2	49.8 ± 7.0	55.2 ± 7.0	68.5 ± 1.0	64.8 ± 1.8	64.5 ± 2.3	80.5 ± 0.8	48.7 ± 2.4	45.4 ± 2.4	74.8 ± 0.5
HIN2vec	24.6 ± 3.5	19.2 ± 7.6	66.4 ± 1.5	23.8 ± 5.5	17.7 ± 13.4	74.6 ± 1.2	41.7 ± 5.3	44.9 ± 10.7	80.1 ± 0.8	35.1 ± 8.3	28.9 ± 8.5	77.8 ± 1.0
GATNE	36.3 ± 8.5	28.5 ± 11.4	67.1 ± 0.6	46.5 ± 15.0	46.0 ± 15.9	74.6 ± 1.2	69.3 ± 3.1	83.9 ± 0.2	54.2 ± 10.7	47.2 ± 11.5	76.2 ± 1.3	76.2 ± 1.3
NSHE	38.3 ± 7.4	29.1 ± 10.2	69.9 ± 0.4	40.0 ± 11.3	39.5 ± 9.4	68.6 ± 1.2	62.5 ± 12.6	61.1 ± 9.1	83.3 ± 0.7	54.3 ± 1.1	48.0 ± 2.6	77.5 ± 0.4
NOH	43.6 ± 1.4	39.4 ± 4.3	67.7 ± 0.5	48.1 ± 3.4	57.2 ± 2.9	76.2 ± 0.5	35.9 ± 3.3	37.7 ± 3.3	72.3 ± 0.5	44.8 ± 12.3	48.0 ± 2.6	73.5 ± 2.2
HeteHG-VAE	51.5 ± 1.4	45.4 ± 1.8	72.7 ± 0.7	48.9 ± 6.1	57.4 ± 5.1	76.7 ± 0.8	35.0 ± 8.5	32.5 ± 10.5	70.3 ± 7.1	54.7 ± 4.0	56.7 ± 3.5	75.4 ± 1.0
HGT	54.8 ± 2.3	53.8 ± 5.7	74.2 ± 1.6	62.0 ± 0.9	60.4 ± 2.1	76.2 ± 2.2	72.1 ± 5.9	73.4 ± 8.0	84.0 ± 0.9	49.7 ± 3.9	48.2 ± 9.7	72.5 ± 1.9
SLICE	11.0 ± 0.4	12.7 ± 0.9	43.8 ± 0.5	11.5 ± 1.0	14.7 ± 2.3	44.1 ± 0.6	10.7 ± 0.7	12.9 ± 1.8	44.1 ± 0.6	9.9 ± 1.4	12.4 ± 0.9	43.1 ± 0.7
SimpleHGN	14.4 ± 2.1	5.8 ± 0.6	46.5 ± 1.1	39.7 ± 4.0	45.8 ± 4.5	59.7 ± 2.3	36.3 ± 7.0	39.4 ± 6.4	58.2 ± 3.9	14.4 ± 2.1	5.8 ± 0.6	46.5 ± 1.1
GraphGPT	14.3 ± 0.0	1.9 ± 0.0	43.7 ± 0.0	14.3 ± 0.0	1.8 ± 0.0	43.7 ± 0.0	14.3 ± 0.0	14.3 ± 0.0	43.7 ± 0.0	14.3 ± 0.0	2.6 ± 0.0	43.7 ± 0.0
HINCHOR w/o CF	89.2 ± 0.8	89.5 ± 0.7	86.4 ± 0.1	88.1 ± 2.8	88.1 ± 2.7	85.3 ± 0.5	88.7 ± 1.2	91.8 ± 1.1	86.8 ± 0.8	80.1 ± 1.1	80.1 ± 1.2	85.7 ± 0.3
HINCHOR	90.1 ± 0.8	90.3 ± 0.7	86.6 ± 0.1	88.6 ± 2.2	89.0 ± 2.2	85.9 ± 0.2	89.4 ± 0.6	92.0 ± 1.1	87.0 ± 0.3	79.0 ± 6.5	78.2 ± 6.6	86.0 ± 0.4

model is HGT, which learns the embeddings by reconstructing the triplet. HINCHOR outperforms HGT, indicating that considering only pairwise relations is not sufficient for predicting higher-order relations. Results on the other two datasets and the comparisons with SLICE and SimpleHGN also validate this observation. In Aminer and IMDB, DeepWalk, GATNE and NSHE show relatively good results. This may be because they preserve relatively rich structural information by applying random walks or network schema.

Among baselines, metapath-based methods such as Metapath2vec and HIN2vec perform less well due to their limited ability to capture direct neighborhood information. Additionally, although link prediction models such as HGT can achieve satisfactory results in pairwise node prediction, the formation of the network does not occur pairwise. Predicting pairwise relations one by one and then combining them may fail to capture the essence of higher-order relations. A single prediction error could lead to the collapse of the entire higher-order relation prediction. The results of hyperedge-based methods NOH and HeteHG-VAE are significantly lower than those of our model. This is because these methods focus on the relations between other nodes and the central node when considering hyperedges, neglecting the relations between non-central nodes. As a result, these methods fail to capture the proximity between non-central nodes, leading to limitations in predicting higher-order relations.

GraphGPT, which integrates the GNN and LLM, fails to predict higher-order relations. When applied as a link predictor, it consistently predicts that every pair of nodes in any higher-order relation test case is connected, resulting in the pattern shown as pattern 8 in Fig. 3. This failure stems from GraphGPT's heavy reliance on textual input — limited to type information in our dataset — making it incapable of adapting to tasks requiring heterogeneous and higher-order relational awareness.

To evaluate the effectiveness of our design of counterfactual learning, we conduct experiments with a variant of HINCHOR where the counterfactual components were removed. This variant is denoted as **HINCHOR w/o CF**. As observed, HINCHOR consistently outperforms the variant in most cases, demonstrating the effectiveness of counterfactual data augmentation. We further evaluate the counterfactual

decoder's ability to predict counterfactual relations by excluding the factual components (denoted as **HINCHOR w/o F**). The results, presented in 5, demonstrate that while predicting counterfactual relations is inherently more challenging, HINCHOR w/o F still performs well. The strong effectiveness of both the factual and counterfactual components forms the foundation for accurately estimating the causal effects between global structures and higher-order relations.

5.3. Causal effect evaluation

To assess whether HINCHOR can effectively learn the causal relationship between the global structure and higher-order relations, we use the Average Treatment Effect (ATE) to evaluate the influence of different treatments on outcomes. We first calculate the observed ATE (ATE_o) by contrasting the observed higher-order relations P_s with created counterfactual higher-order relations P_{s_c} that have opposite treatments. Then we calculate the estimated ATE (ATE_e) by comparing the observed higher-order relations and predicted counterfactual higher-order relations $\hat{P}_{s_c}$. These two metrics allow us to further investigate the relationship between the estimated ATE and the prediction performance. Formally, they are defined as:

$$ATE_o = \frac{1}{|P|} \sum_{p \in P} \frac{1}{|S^p|} \sum_{s \in S^p} \left\{ T_s \cdot (P_s - P_{s_c}) + (1 - T_s) \cdot (P_{s_c} - P_s) \right\},$$

$$ATE_e = \frac{1}{|P|} \sum_{p \in P} \frac{1}{|S^p|} \sum_{s \in S^p} \left\{ T_s \cdot (P_s - \hat{P}_{s_c}) + (1 - T_s) \cdot (\hat{P}_{s_c} - P_s) \right\},$$

where P represents the set of all possible patterns involving a certain number of nodes, and $S^p = \{s | P_s = p \wedge s \in S\}$.

We first select a couple of community detection algorithms to derive the treatment variables, including SBM (Karrer & Newman, 2011), K-Core algorithm (Bader & Hogue, 2003), Spectral Clustering (SC) (Ng, Jordan, & Weiss, 2001), Louvain algorithm (Blondel, Guillaume, Lambiotte, & Lefebvre, 2008) and Hierarchical Clustering (HC) (Ward, 1963). We employ ATE evaluation on DouBan and DBLP and report the results in Table 6, sorted by descending Micro-F1 scores. We calculate Kendall's ranking coefficient (Abdi, 2007) between the ATE_o and ATE_e ,

Table 5

Factual and counterfactual prediction on Douban and DBLP datasets.

Dataset	Methods	Macro-F1	Micro-F1	Precision	Recall	AUC	NMI
DouBan	HINCHOR	0.885 $\pm$ 0.024	0.887 $\pm$ 0.023	0.890 $\pm$ 0.022	0.886 $\pm$ 0.022	0.859 $\pm$ 0.002	0.809 $\pm$ 0.022
	HINCHOR w/o CF	0.879 $\pm$ 0.030	0.883 $\pm$ 0.029	0.881 $\pm$ 0.027	0.881 $\pm$ 0.028	0.853 $\pm$ 0.005	0.804 $\pm$ 0.027
	HINCHOR w/o F	0.854 $\pm$ 0.035	0.860 $\pm$ 0.031	0.866 $\pm$ 0.033	0.859 $\pm$ 0.030	0.855 $\pm$ 0.003	0.784 $\pm$ 0.034
DBLP	HINCHOR	0.900 $\pm$ 0.008	0.901 $\pm$ 0.008	0.903 $\pm$ 0.007	0.901 $\pm$ 0.008	0.866 $\pm$ 0.001	0.787 $\pm$ 0.014
	HINCHOR w/o CF	0.891 $\pm$ 0.008	0.892 $\pm$ 0.008	0.895 $\pm$ 0.007	0.892 $\pm$ 0.008	0.864 $\pm$ 0.001	0.772 $\pm$ 0.010
	HINCHOR w/o F	0.730 $\pm$ 0.060	0.740 $\pm$ 0.050	0.763 $\pm$ 0.059	0.740 $\pm$ 0.050	0.835 $\pm$ 0.013	0.617 $\pm$ 0.030

Table 6

Experiment results with different community treatments on Douban and DBLP datasets (sorted based on Micro-F1).

Dataset	Methods	Micro-F1	ATE _o	ATE _e
DouBan	SBM	0.887 $\pm$ 0.023	-0.374	-0.431 $\pm$ 0.028
	SC	0.801 $\pm$ 0.029	-0.126	-0.161 $\pm$ 0.012
	HC	0.792 $\pm$ 0.072	-0.207	-0.293 $\pm$ 0.029
	Louvain	0.785 $\pm$ 0.076	-0.023	-0.032 $\pm$ 0.027
	K-Core	0.634 $\pm$ 0.079	-0.016	-0.125 $\pm$ 0.019
DBLP	HC	0.903 $\pm$ 0.008	-0.036	-0.160 $\pm$ 0.025
	SC	0.902 $\pm$ 0.011	-0.123	-0.236 $\pm$ 0.014
	SBM	0.901 $\pm$ 0.008	-0.312	-0.407 $\pm$ 0.021
	K-Core	0.895 $\pm$ 0.009	-0.288	-0.379 $\pm$ 0.014
	Louvain	0.867 $\pm$ 0.026	-0.311	-0.413 $\pm$ 0.021

Table 7

Experiment results with different closeness treatments on Douban datasets.

Threshold	Micro-F1	ATE _o	ATE _e
0.064	0.786 $\pm$ 0.008	0.170	0.114 $\pm$ 0.006
0.032	0.755 $\pm$ 0.047	0.035	0.056 $\pm$ 0.004
0.024	0.803 $\pm$ 0.035	-0.008	0.010 $\pm$ 0.009
0.016	0.806 $\pm$ 0.046	-0.010	-0.001 $\pm$ 0.006
0.008	0.805 $\pm$ 0.046	-0.007	0.005 $\pm$ 0.017

resulting in the value of 0.8 both for DouBan and DBLP, indicating a strong correlation. This demonstrates that our model effectively learns the causal relationship. Besides, we observe that ATE_o and ATE_e are both roughly correlated with Micro-F1. This could be attributed to the causal relationship between the graph structure and higher-order relations, making it easier for the model to capture the fundamental factors influencing prediction. This reduced ambiguity contributes to the model's performance improvement. Consequently, selecting an appropriate treatment-driven method before training the model can lead to better learning of higher-order relation prediction.

We additionally select closeness centrality (the mean centrality of nodes within higher-order relations) as global features. We use thresholds to differentiate between different treatments, where varying thresholds may lead to different treatments for higher-order relations. The results are provided in Table 7. We calculated the Kendall's rank correlation coefficient between ATE_o and ATE_e, which is 0.8, indicating a strong correlation. By using both community and centrality as global structural factors, we show that our model effectively learns causal relationships, and our framework can be used to evaluate factors influencing the prediction of higher-order relations.

5.4. Ablation experiment

To validate the effectiveness of the higher-order structure encoder, we conducted the following ablation experiments.

We train 5 variants of HINCHOR by modifying the biased random walk to an unbiased setting (**HINCHOR w/o BRW**), removing node anonymization (**HINCHOR w/o NA**), replacing the LSTM module with a GRU module (**HINCHOR-GRU**) and a self-attention module (**HINCHOR-ATT**) respectively, and replacing the entire encoder with

a GNN (**HINCHOR-GNN**). The results of the higher-order relation prediction on the DouBan are presented in Table 8. As expected, removing the biased random walk will cause a certain degree of decline in the effect. Removing node anonymization or replacing the encoder with GNN will greatly reduce the effect of the model and lose the ability to capture the proximity within heterogeneous higher-order relations. Using the GRU module also results in a certain degree of performance degradation, as the LSTM we employ is better at capturing the implicit and long-term dependencies within the sequence. The variant using the self-attention module completely fails, predicting all test cases as pattern 1 in Fig. 3. This failure occurs because the input sequences contain a large number of zero elements, making it challenging for the dot-product-based attention mechanism to effectively learn the relationships between different positions. All these results prove the effectiveness of our heterogeneous higher-order structure encoder.

We further conduct experiments using three different types of random walks to validate the capability of our encoding strategy: Biased Random Walk (BRW), Random Walk (RW), and Random Walk with Restart (RWR). Each type of walk was paired with different encoding methods: Anonymous Encoding (NA) and Network Embedding Encoding (NE, here we use Metapath2vec). As shown in the Table 9, NE encoding consistently performed worse than NA encoding across all types of random walks, indicating that our anonymous encoding is more effective at capturing context and proximity. Among the different random walk methods, RWR outperformed the others, suggesting that RWR is more effective at exploring the local structure near the starting node through periodic restarts.

Combined with the results in Section 5.2, we find that random walk-based methods such as DeepWalk, usually show promising results. Similarly, our model utilizes random walks to model context, showcasing the effectiveness of random walks in capturing higher-order proximity. Furthermore, biasing random walks contributes to improved performance. Metapath2vec and HIN2vec have shown mediocre results, mainly due to their over-reliance on meta-paths, limiting their ability to capture direct neighbor proximity. Although GATNE learns node embeddings by aggregating neighbor information from different edge types, it only achieves moderate results on the DBLP and DouBan datasets. Similarly, there is a notable decrease in performance where the encoder of HINCHOR is replaced by a HGNN. This suggests that while message-passing methods can capture some proximity, they may suffer from over-smoothing, leading to poor embedding quality and affecting relation inference.

5.5. Choices of fixed nodes

When searching for counterfactuals, we select central nodes that are connected to all other types of nodes in the network. To explore the impact of this strategy, we replace the fixed node type with one of the other types and implement the prediction task on Douban dataset. The results are in the Table 10. We can see that using other nodes as the fixed node results in a certain degree of performance decline. This proves that the counterfactual searched by the fixed central nodes is more meaningful since the central nodes can usually be considered as the unique identifier of the events forming a network.

Table 8

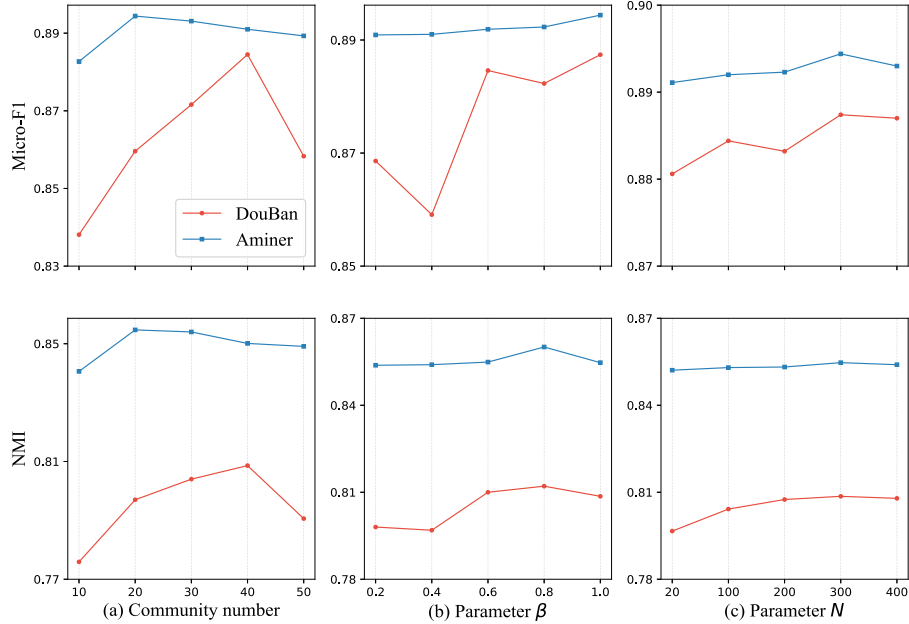
Ablation Experiment results on the DouBan datasets for the higher-order relation prediction task.

Encoder	AUC	NMI	Macro-F1	Micro-F1	Precision	Recall
HINCHOR	0.859 $\pm$ 0.002	0.809 $\pm$ 0.022	0.885 $\pm$ 0.024	0.887 $\pm$ 0.023	0.890 $\pm$ 0.022	0.886 $\pm$ 0.022
HINCHOR w/o BRW	0.857 $\pm$ 0.006	0.804 $\pm$ 0.015	0.879 $\pm$ 0.032	0.883 $\pm$ 0.030	0.889 $\pm$ 0.024	0.882 $\pm$ 0.029
HINCHOR w/o NA	0.789 $\pm$ 0.015	0.693 $\pm$ 0.008	0.603 $\pm$ 0.023	0.632 $\pm$ 0.029	0.635 $\pm$ 0.0232	0.628 $\pm$ 0.028
HINCHOR-GRU	0.856 $\pm$ 0.0026	0.769 $\pm$ 0.043	0.830 $\pm$ 0.083	0.840 $\pm$ 0.066	0.867 $\pm$ 0.026	0.838 $\pm$ 0.067
HINCHOR-ATT	0.438 $\pm$ 0.000	0.000 $\pm$ 0.000	0.363 $\pm$ 0.000	0.146 $\pm$ 0.000	0.021 $\pm$ 0.000	0.143 $\pm$ 0.000
HINCHOR-GNN	0.747 $\pm$ 0.024	0.492 $\pm$ 0.200	0.483 $\pm$ 0.195	0.533 $\pm$ 0.178	0.512 $\pm$ 0.149	0.531 $\pm$ 0.180

Table 9

Different random walk coding strategy results on the DouBan datasets for the higher-order relation prediction task.

Methods	AUC	NMI	Macro-F1	Micro-F1	Precision	Recall
BRW-NA	0.859 $\pm$ 0.002	0.809 $\pm$ 0.022	0.885 $\pm$ 0.024	0.887 $\pm$ 0.023	0.890 $\pm$ 0.022	0.886 $\pm$ 0.022
BRW-NE	0.789 $\pm$ 0.015	0.693 $\pm$ 0.008	0.603 $\pm$ 0.023	0.632 $\pm$ 0.029	0.635 $\pm$ 0.0232	0.628 $\pm$ 0.028
RW-NA	0.857 $\pm$ 0.006	0.804 $\pm$ 0.015	0.879 $\pm$ 0.032	0.883 $\pm$ 0.030	0.889 $\pm$ 0.024	0.882 $\pm$ 0.029
RW-NE	0.777 $\pm$ 0.011	0.664 $\pm$ 0.003	0.602 $\pm$ 0.002	0.632 $\pm$ 0.006	0.607 $\pm$ 0.004	0.639 $\pm$ 0.010
RWR-NA	0.859 $\pm$ 0.003	0.825 $\pm$ 0.017	0.896 $\pm$ 0.022	0.898 $\pm$ 0.022	0.901 $\pm$ 0.022	0.896 $\pm$ 0.022
RWR-NE	0.788 $\pm$ 0.001	0.686 $\pm$ 0.017	0.614 $\pm$ 0.028	0.640 $\pm$ 0.011	0.634 $\pm$ 0.038	0.640 $\pm$ 0.010

**Fig. 8.** Parameter impact on Douban and Aminer.**Table 10**

Experiment results based on different central nodes on the DouBan dataset for the higher-order relation prediction task.

	Macro-F1	Micro-F1	NMI	Precision	Recall	AUC
Book	0.885 $\pm$ 0.024	0.887 $\pm$ 0.023	0.809 $\pm$ 0.022	0.890 $\pm$ 0.022	0.886 $\pm$ 0.022	0.859 $\pm$ 0.002
Author	0.843 $\pm$ 0.037	0.852 $\pm$ 0.028	0.790 $\pm$ 0.022	0.862 $\pm$ 0.023	0.851 $\pm$ 0.027	0.857 $\pm$ 0.003
Publisher	0.883 $\pm$ 0.019	0.886 $\pm$ 0.018	0.801 $\pm$ 0.017	0.889 $\pm$ 0.018	0.884 $\pm$ 0.018	0.858 $\pm$ 0.004

5.6. Parameter analysis

Here we investigate the impact of several important parameters. We conduct experiments by changing the value of a single parameter while keeping all other parameters at their default settings. As shown in Fig. 8, the prediction results exhibit an initial improvement followed by a decline as the number of communities increases. The optimal performance is achieved with 40 partitions on DouBan and 20 partitions on Aminer. This demonstrates the significance of selecting an appropriate value for different datasets. A larger value of the balance weight β (such as 0.8-1) makes the positive impact of counterfactual learning on prediction results more significant. The prediction performance initially

improves as N increases, which means that sampling more random walks for each node captures more context and leads to better model performance. However, the performance stabilizes after reaching a certain point, so we choose $N = 300$ to balance cost and performance. Fig. 9 shows the HINCHOR performance under different combinations of the bias parameters b and q . On the DouBan and Aminer datasets, the best prediction performance is achieved when $b = 1$ and q is set to 2 and 3, respectively. With the increase of b , the performance rapidly declines, because smaller values of b confine the random walks to the vicinity of the source node, which is more conducive to capturing multi-node proximity. However, as the value of q decreases, the model

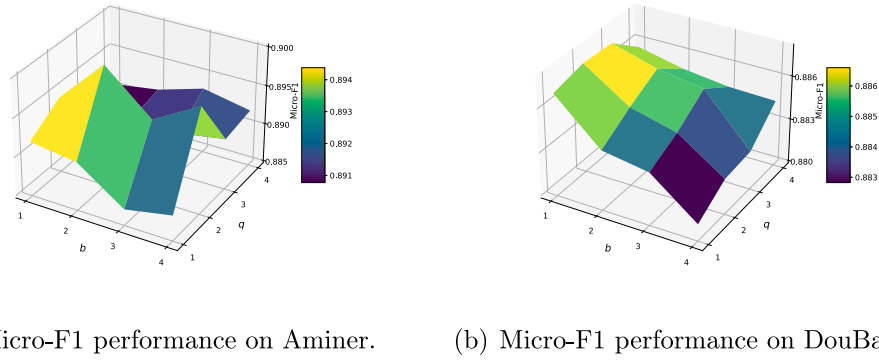


Fig. 9. Performance of HINCHOR on Aminer and DouBan w.r.t different combinations of b and q .

tends to favor deeper walks, which is not conducive to capturing local information.

6. Conclusion

In this paper, we have introduced HINCHOR, a framework to predict higher-order relations on HINs. HINCHOR incorporates an encoder to capture the proximity of higher-order relations, taking both structural and type information into account. By introducing counterfactual learning, we explore the causal relationship between graph structure and relation state, marking the first attempt to apply counterfactual learning to higher-order relation prediction. We construct four real-world benchmark datasets for evaluation. HINCHOR achieves superior performance over state-of-the-art baselines across seven metrics on high-order relationship prediction tasks. Notably, our approach improves macro-F1 and micro-F1 scores 25.0%–38.1% and 20.5%–35.3%, respectively.

Looking ahead, there are several avenues for future expansion in our work. In HINs, various factors can influence higher-order relation prediction beyond just community information. Additionally, we plan to extend our research to the temporal networks, which presents a new challenge in handling temporal dependencies.

CRediT authorship contribution statement

Xuan Guo: Writing – original draft, Methodology, Investigation, Conceptualization. **Jie Li:** Validation, Investigation. **Pengfei Jiao:** Writing – review & editing. **Wang Zhang:** Data curation. **Tianpeng Li:** Writing – review & editing. **Wenjun Wang:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The link of data and code has been provided in the manuscript.

References

- Abdi, H. (2007). The kendall rank correlation coefficient. In *Encyclopedia of measurement and statistics* (pp. 508–510). Thousand Oaks, CA: Sage.
- Abbate, C., & Bonchi, F. (2021). Counterfactual graphs for explainable classification of brain networks. In *Proceedings of the 27th ACM SIGKDD conference on knowledge discovery & data mining* (pp. 2495–2504).
- Agarwal, C., Lakkaraju, H., & Zitnik, M. (2021). Towards a unified framework for fair and stable graph representation learning. In *Uncertainty in artificial intelligence* (pp. 2114–2124). PMLR.
- Alon, U. (2007). Network motifs: Theory and experimental approaches. *Nature Reviews. Genetics*, 8(6), 450–461.
- Bader, G. D., & Hogue, C. W. (2003). An automated method for finding molecular complexes in large protein interaction networks. *BMC Bioinformatics*, 4(1), 1–27.
- Benson, A. R., Abebe, R., Schaub, M. T., Jadbabaie, A., & Kleinberg, J. (2018). Simplicial closure and higher-order link prediction. *Proceedings of the National Academy of Sciences*, 115(48), E11221–E11230.
- Benson, A. R., Gleich, D. F., & Higham, D. J. (2021). Higher-order network analysis takes off, fueled by classical ideas and new data. arXiv preprint arXiv:2103.05031.
- Benson, A. R., Gleich, D. F., & Leskovec, J. (2016). Higher-order organization of complex networks. *Science*, 353(6295), 163–166.
- Besta, M., Grob, R., Miglioni, C., Bernold, N., Kwasniewski, G., Gjini, G., et al. (2022). Motif prediction with graph neural networks. In *Proceedings of the 28th ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 35–45).
- Blondel, V. D., Guillaume, J.-L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2008(10), Article P10008.
- Cen, Y., Zou, X., Zhang, J., Yang, H., Zhou, J., & Tang, J. (2019). Representation learning for attributed multiplex heterogeneous network. In *Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining* (pp. 1358–1368).
- Ceria, A., & Wang, H. (2023). Temporal-topological properties of higher-order evolving networks. *Scientific Reports*, 13(1), 5885.
- Chang, H., Cai, J., & Li, J. (2023). Knowledge graph completion with counterfactual augmentation. In *Proceedings of the web conference 2023* (pp. 2611–2620).
- Chen, Z., Silvestri, F., Wang, J., Zhang, Y., Huang, Z., Ahn, H., et al. (2022). Grease: Generate factual and counterfactual explanations for gnn-based recommendations. arXiv preprint arXiv:2208.04222.
- Chung, C., & Whang, J. J. (2021). Knowledge graph embedding via metagraph learning. In *Proceedings of the 44th international ACM SIGIR conference on research and development in information retrieval* (pp. 2212–2216).
- Dareddy, M. R., Das, M., & Yang, H. (2019). Motif2vec: Motif aware node representation learning for heterogeneous networks. In *2019 IEEE international conference on big data* (pp. 1052–1059). IEEE.
- Dong, Y., Chawla, N. V., & Swami, A. (2017). Metapath2vec: Scalable representation learning for heterogeneous networks. In *Proceedings of the 23rd ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 135–144).
- Fan, H., Zhang, F., Wei, Y., Li, Z., Zou, C., Gao, Y., et al. (2021). Heterogeneous hypergraph variational autoencoder for link prediction. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(8), 4125–4138.
- Fu, T.-y., Lee, W.-C., & Lei, Z. (2017). Hin2vec: Explore meta-paths in heterogeneous information networks for representation learning. In *Proceedings of the 2017 ACM international conference on information & knowledge management* (pp. 1797–1806).
- Fu, X., Zhang, J., Meng, Z., & King, I. (2020). Magnn: Metapath aggregated graph neural network for heterogeneous graph embedding. In *Proceedings of the web conference 2020* (pp. 2331–2341).
- Graves, A., & Graves, A. (2012). Long short-term memory. In *Supervised sequence labelling with recurrent neural networks* (pp. 37–45).
- Grover, A., & Leskovec, J. (2016). Node2vec: Scalable feature learning for networks. In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery & data mining* (pp. 855–864).

- Guan, M., Cai, X., Shang, J., Hao, F., Liu, D., Jiao, X., et al. (2023). Hmsg: Heterogeneous graph neural network based on metapath subgraph learning. *Knowledge-Based Systems*, 279, Article 110930.
- Guo, X., Jiao, P., Zhang, W., Pan, T., Jia, M., Shi, D., et al. (2024). Representation learning on heterostructures via heterogeneous anonymous walks. *IEEE Transactions on Neural Networks and Learning Systems*, 35(7), 9538–9552.
- Guo, Z., Xiao, T., Aggarwal, C., Liu, H., & Wang, S. (2023). Counterfactual learning on graphs: A survey. *arXiv preprint arXiv:2304.01391*.
- Hao, P., Qian, Z., Wang, S., & Bai, C. (2023). Community aware graph embedding learning for item recommendation. *World Wide Web*, 26(6), 4093–4108.
- Hu, Z., Dong, Y., Wang, K., & Sun, Y. (2020). Heterogeneous graph transformer. In *Proceedings of the web conference 2020* (pp. 2704–2710).
- Hu, B., Fang, Y., & Shi, C. (2019). Adversarial learning on heterogeneous information networks. In *Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining* (pp. 120–129).
- Karrer, B., & Newman, M. E. (2011). Stochastic blockmodels and community structure in networks. *Physical Review E*, 83(1), Article 016107.
- Kingma, D. P., & Ba, J. (2015). Adam: A method for stochastic optimization. In *3rd international conference on learning representations*.
- Lahiri, M., & Berger-Wolf, T. Y. (2007). Structure prediction in temporal networks using frequent subgraphs. In *2007 IEEE symposium on computational intelligence and data mining* (pp. 35–42). IEEE.
- Lambiotte, R., Rosvall, M., & Scholtes, I. (2019). From networks to optimal higher-order models of complex systems. *Nature Physics*, 15(4), 313–320.
- Li, W.-Z., Huang, L., Wang, C.-D., & Ye, Y.-X. (2021). Stargat: Star-shaped hierarchical graph attentional network for heterogeneous network representation learning. In *2021 IEEE international conference on data mining* (pp. 1198–1203). IEEE.
- Li, Z., Jian, X., Wang, Y., & Chen, L. (2022). Cc-gnn: A community and contraction-based graph neural network. In *2022 IEEE international conference on data mining* (pp. 231–240). IEEE.
- Li, Y., Yu, R., Shahabi, C., & Liu, Y. (2018). Diffusion convolutional recurrent neural network: Data-driven traffic forecasting. In *6th international conference on learning representations*.
- Ling, C., Yang, C., & Zhao, L. (2023). Motif-guided heterogeneous graph deep generation. *Knowledge and Information Systems*, 65(7), 3099–3124.
- Liu, C., Gao, C., Jin, D., & Li, Y. (2021). Improving location recommendation with urban knowledge graph. *arXiv preprint arXiv:2111.01013*.
- Liu, Y., Ma, J., & Li, P. (2022). Neural predicting higher-order patterns in temporal networks. In *Proceedings of the web conference 2022* (pp. 1340–1351).
- Lu, Y., Fang, Y., & Shi, C. (2020). Meta-learning on heterogeneous information networks for cold-start recommendation. In *Proceedings of the 26th ACM SIGKDD international conference on knowledge discovery & data mining* (pp. 1563–1573).
- Lu, Y., Gao, M., Liu, H., Liu, Z., Yu, W., Li, X., et al. (2023). Neighborhood overlap-aware heterogeneous hypergraph neural network for link prediction. *Pattern Recognition*, 144, Article 109818.
- Lu, Y., Shi, C., Hu, L., & Liu, Z. (2019). Relation structure-aware heterogeneous information network embedding. In *Proceedings of the thirty-third AAAI conference on artificial intelligence*, vol. 33 (pp. 4456–4463).
- Lucic, A., Ter Hoeve, M. A., Tolomei, G., De Rijke, M., & Silvestri, F. (2022). Cf-gnnexplainer: Counterfactual explanations for graph neural networks. In *International conference on artificial intelligence and statistics* (pp. 4499–4511). PMLR.
- Lv, Q., Ding, M., Liu, Q., Chen, Y., Feng, W., He, S., et al. (2021). Are we really making much progress? Revisiting, benchmarking and refining heterogeneous graph neural networks. In *Proceedings of the 27th ACM SIGKDD international conference on knowledge discovery & data mining* (pp. 1150–1160).
- Ma, X., Tan, S., Xie, X., Zhong, X., & Deng, J. (2022). Joint multi-label learning and feature extraction for temporal link prediction. *Pattern Recognition*, 121, Article 108216.
- Ma, X., Zhang, B., Ma, C., & Ma, Z. (2020). Co-regularized nonnegative matrix factorization for evolving community detection in dynamic networks. *Information Sciences*, 528, 265–279.
- Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., & Galstyan, A. (2021). A survey on bias and fairness in machine learning. *ACM Computing Surveys (CSUR)*, 54(6), 1–35.
- Milo, R., Shen-Orr, S., Itzkovitz, S., Kashtan, N., Chklovskii, D., & Alon, U. (2002). Network motifs: Simple building blocks of complex networks. *Science*, 298(5594), 824–827.
- Nassar, H., Benson, A. R., & Gleich, D. F. (2020). Neighborhood and pagerank methods for pairwise link prediction. *Social Network Analysis and Mining*, 10, 1–13.
- Navlakha, S., & Kingsford, C. (2010). The power of protein interaction networks for associating genes with diseases. *Bioinformatics*, 26(8), 1057–1063.
- Newman, M. E., Watts, D. J., & Strogatz, S. H. (2002). Random graph models of social networks. *Proceedings of the National Academy of Sciences*, 99(suppl_1), 2566–2572.
- Ng, A., Jordan, M., & Weiss, Y. (2001). On spectral clustering: Analysis and an algorithm. *Advances in Neural Information Processing Systems*, 14, 849–856.
- Nguyen, T.-K., Liu, Z., & Fang, Y. (2023). Link prediction on latent heterogeneous graphs. In *Proceedings of the web conference 2023* (pp. 263–273).
- Perozzi, B., Al-Rfou, R., & Skiena, S. (2014). Deepwalk: Online learning of social representations. In *Proceedings of the 20th ACM SIGKDD international conference on knowledge discovery & data mining* (pp. 701–710).
- Phan, A. V., Le Nguyen, M., Nguyen, Y. L. H., & Bui, L. T. (2018). Dgcnn: A convolutional neural network over large-scale labeled graphs. *Neural Networks*, 108, 533–543.
- Rossi, R. A., Ahmed, N. K., Carranza, A., Arbour, D., Rao, A., Kim, S., et al. (2020). Heterogeneous graphlets. *ACM Transactions on Knowledge Discovery from Data (TKDD)*, 15(1), 1–43.
- Rossi, R. A., Jin, D., Kim, S., Ahmed, N. K., Koutra, D., & Lee, J. B. (2020). On proximity and structural role-based embeddings in networks: Misconceptions, techniques, and applications. *ACM Transactions on Knowledge Discovery from Data (TKDD)*, 14(5), 1–37.
- Tang, J., Yang, Y., Wei, W., Shi, L., Su, L., Cheng, S., et al. (2024). Graphpnt: Graph instruction tuning for large language models. In *Proceedings of the 47th international ACM SIGIR conference on research and development in information retrieval* (pp. 491–500).
- Tian, Q., Zhang, W., Jiao, P., Zhong, K., Wu, N., & Pan, L. (2022). Integrating higher-order features for structural role discovery. In *International conference on mobile computing, applications, and services* (pp. 244–258). Springer.
- Wang, P., Agarwal, K., Ham, C., Choudhury, S., & Reddy, C. K. (2021). Self-supervised learning of contextual embeddings for link prediction in heterogeneous networks. In *Proceedings of the web conference 2021* (pp. 2946–2957).
- Wang, Y., Chang, Y.-Y., Liu, Y., Leskovec, J., & Li, P. (2021). Inductive representation learning in temporal networks via causal anonymous walks. In *9th international conference on learning representations*.
- Wang, W., Feng, F., He, X., Zhang, H., & Chua, T.-S. (2021). Clicks can be cheating: Counterfactual recommendation for mitigating clickbait issue. In *Proceedings of the 44th international ACM SIGIR conference on research and development in information retrieval* (pp. 1288–1297).
- Wang, W., Liu, J., Tang, T., Tuorab, S., Xia, F., Gong, Z., et al. (2020). Attributed collaboration network embedding for academic relationship mining. *ACM Transactions on the Web (TWEB)*, 15(1), 1–20.
- Ward, J. H., Jr. (1963). Hierarchical grouping to optimize an objective function. *Journal of the American Statistical Association*, 58(301), 236–244.
- Wu, W., & Ma, X. (2022). Network-based structural learning nonnegative matrix factorization algorithm for clustering of scrna-seq data. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 20(1), 566–575.
- Xu, Y., Feng, Z., Xing, M., Wu, H., Chen, S., Xue, X., et al. (2023). Metapath-guided multi-headed attention networks for trust prediction in heterogeneous social networks. *Knowledge-Based Systems*, 282, Article 111119.
- Yadati, N., Nitin, V., Nimishakavi, M., Yadav, P., Louis, A., & Talukdar, P. (2020). Nhp: Neural hypergraph link prediction. In *Proceedings of the 29th ACM international conference on information & knowledge management* (pp. 1705–1714).
- Yang, H., Li, L., Zhang, L., Tang, J., & Chen, Z. (2021). Phgnn: Position-aware graph neural network for heterogeneous graph embedding. In *2021 international joint conference on neural networks* (pp. 1–8). IEEE.
- Yang, C., Sun, M., Liu, Z., & Tu, C. (2017). Fast network embedding enhancement via high order proximity approximation. In *Proceedings of the 26th International Joint Conference on Artificial Intelligence* (pp. 3894–3900).
- Yang, C., Xiao, Y., Zhang, Y., Sun, Y., & Han, J. (2020). Heterogeneous network representation learning: A unified framework with survey and benchmark. *IEEE Transactions on Knowledge and Data Engineering*, 34(10), 4854–4873.
- Yao, L., Chu, Z., Li, S., Li, Y., Gao, J., & Zhang, A. (2021). A survey on causal inference. *ACM Transactions on Knowledge Discovery from Data (TKDD)*, 15(5), 1–46.
- Yu, L., Luo, Z., Liu, H., Lin, D., Li, H., & Deng, Y. (2022). Triplere: Knowledge graph embeddings via triple relation vectors. *arXiv preprint arXiv:2209.08271*.
- Zhang, M., & Chen, Y. (2018). Link prediction based on graph neural networks. *Advances in Neural Information Processing Systems*, 31, 5171–5181.
- Zhang, W., Fang, Y., Liu, Z., Wu, M., & Zhang, X. (2020). mg2vec: Learning relationship-preserving heterogeneous graph representations via metagraph embedding. *IEEE Transactions on Knowledge and Data Engineering*, 34(3), 1317–1329.
- Zhang, W., Guo, X., Pan, T., Liu, C., Jiao, P., Pan, L., et al. (2021). Role-oriented network embedding based on adversarial learning between higher-order and local features. In *Proceedings of the 30th ACM international conference on information & knowledge management* (pp. 3632–3636).
- Zhang, J., Luxton-Reilly, A., Denny, P., & Whalley, J. (2021). Scientific collaboration network analysis for computing education conferences. In *Proceedings of the 26th ACM conference on innovation and technology in computer science education*, vol. 1 (pp. 582–588).
- Zhang, J., Shi, X., Xie, J., Ma, H., King, I., & Yeung, D.-Y. (2018). Gaan: Gated attention networks for learning on large and spatiotemporal graphs. In *Proceedings of the thirty-fourth conference on uncertainty in artificial intelligence*.
- Zhang, J., Shi, X., Zhao, S., & King, I. (2019). Star-gcn: Stacked and reconstructed graph convolutional networks for recommender systems. In *Proceedings of the twenty-eighth international joint conference on artificial intelligence* (pp. 4264–4270).
- Zhang, X., Yang, Q., & Xu, D. (2022). Trans: Transition-based knowledge graph embedding with synthetic relation representation. In *Findings of the association for computational linguistics 2022* (pp. 1202–1208).
- Zhao, T., Liu, G., Wang, D., Yu, W., & Jiang, M. (2022). Learning from counterfactual links for link prediction. In *International conference on machine learning* (pp. 26911–26926). PMLR.
- Zhao, J., Wang, X., Shi, C., Liu, Z., & Ye, Y. (2020). Network schema preserving heterogeneous information network embedding. In *Proceedings of the twenty-ninth international joint conference on artificial intelligence* (pp. 1366–1372).